PRODUCER GUIDE

RatioBos 1.0.0

# The *cheapest* ration for your cattle, from the pen to the profit

A practical guide to formulating, mixing, costing and measuring the footprint of your feedlot ration with RatioBos, explained through one real case from Jalisco, start to finish.

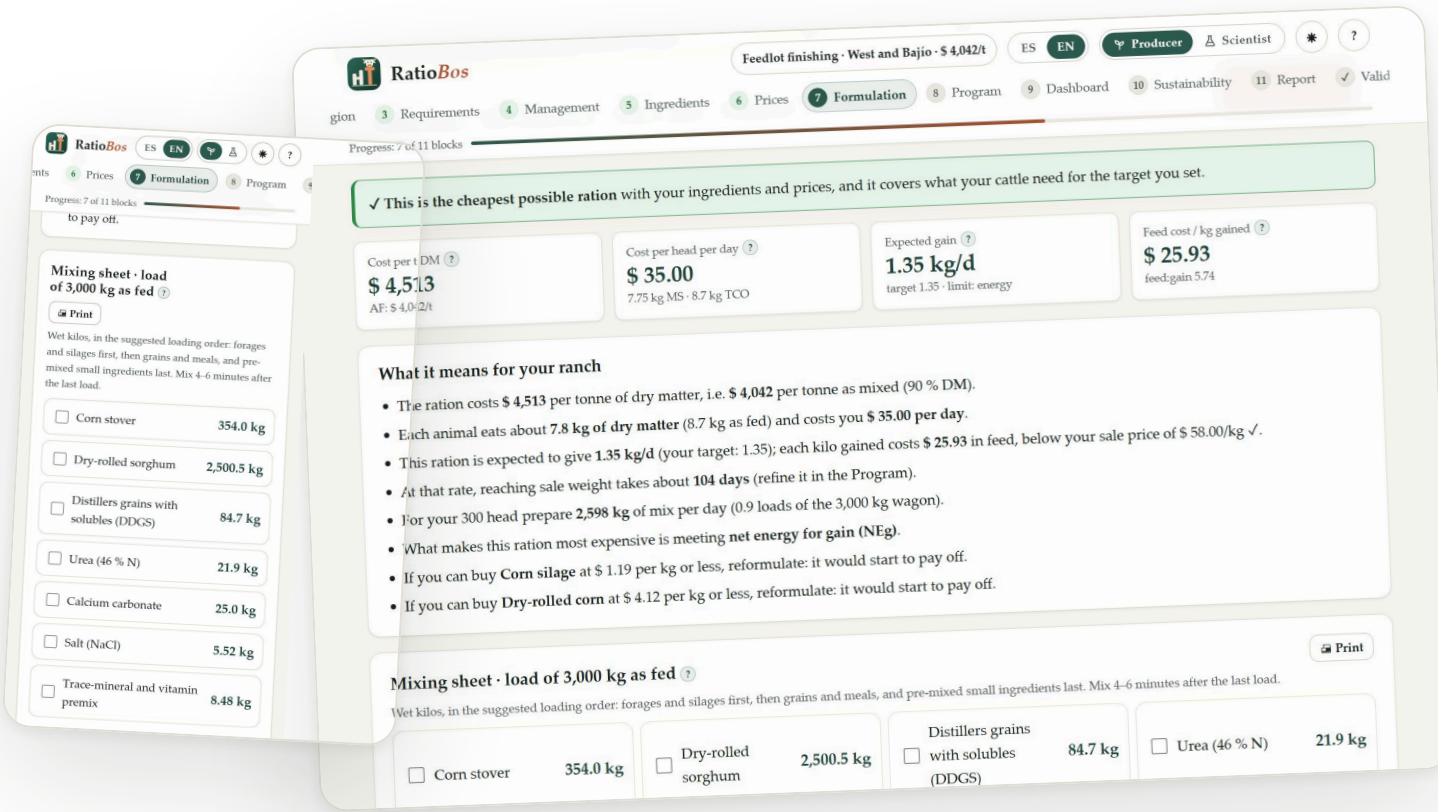
Open the app: **hectortecumshe-ai.github.io/RatioBos**

**1st edition** · October 2026 · covers RatioBos 1.0.0

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*There is a companion guide for research and teaching: the Scientist Guide.*



BEFORE READING

# How to use this guide

Who it is for, how it is organised and what each sign on the page means.

This guide is for whoever decides what the cattle on a ranch or in a feedlot eat: the producer, the mixer-wagon operator, the veterinarian or nutritionist who advises several operations. You need no coding or maths; you need to know your cattle.

It follows the app's order: one chapter for each block in the top bar, 1 to 11, each with its own colour. Every example uses **a single case**, a 300-bull feedlot in Jalisco, so the numbers in one chapter are the ones that feed the next. Read it start to finish and you will have formulated, mixed, costed and footprinted a real ration.

## Conventions

| WRITTEN AS | MEANS |
|---|---|
| ⚙ Formulate | An app button or field, with its text exactly as on screen. |
| Report › Save project | A path: open the first item, then the next. |
| 380 | A value you type as shown. |
| 1 | A marker on a screenshot; it is explained below with the same number. |
| **MS · TCO** | Dry matter · as fed (the wet feed, as the mixer scale weighs it). The app writes DM and AF in English. |

# The five boxes

◆ **IN THE PEN**

The worked case, step by step, with the real numbers the app returns.

✓ **TIP**

Shortcuts and good practice that save time or money.

! **CAUTION**

Common mistakes that can cost money or animal health.

? **WHY?**

The reason behind a recommendation, in a few lines. You can skip it.

→ **DECISION RULE**

A short table: if you see this, do that.

✓ **TWO GUIDES, TWO VIEWS**

RatioBos has a **producer view** and a **scientist view** (button in the top bar). This guide covers the producer view. The equations, the linear model, the validation and the research export are in the Scientist Guide.

GETTING STARTED

# 0 Open the app and meet the screen



RatioBos is a web page that runs on its own in your browser. Nothing to install, no account, and your data goes nowhere.

## 0.1 Opening RatioBos

Go to **hectortecumshe-ai.github.io/RatioBos** from a computer, tablet or phone, or scan the QR code on the last page. The first time it downloads a single file of under half a megabyte; after that it works without internet if you leave the tab open, and you can save the page (Ctrl + S) to use it on the ranch with no signal.

## 0.2 The home screen

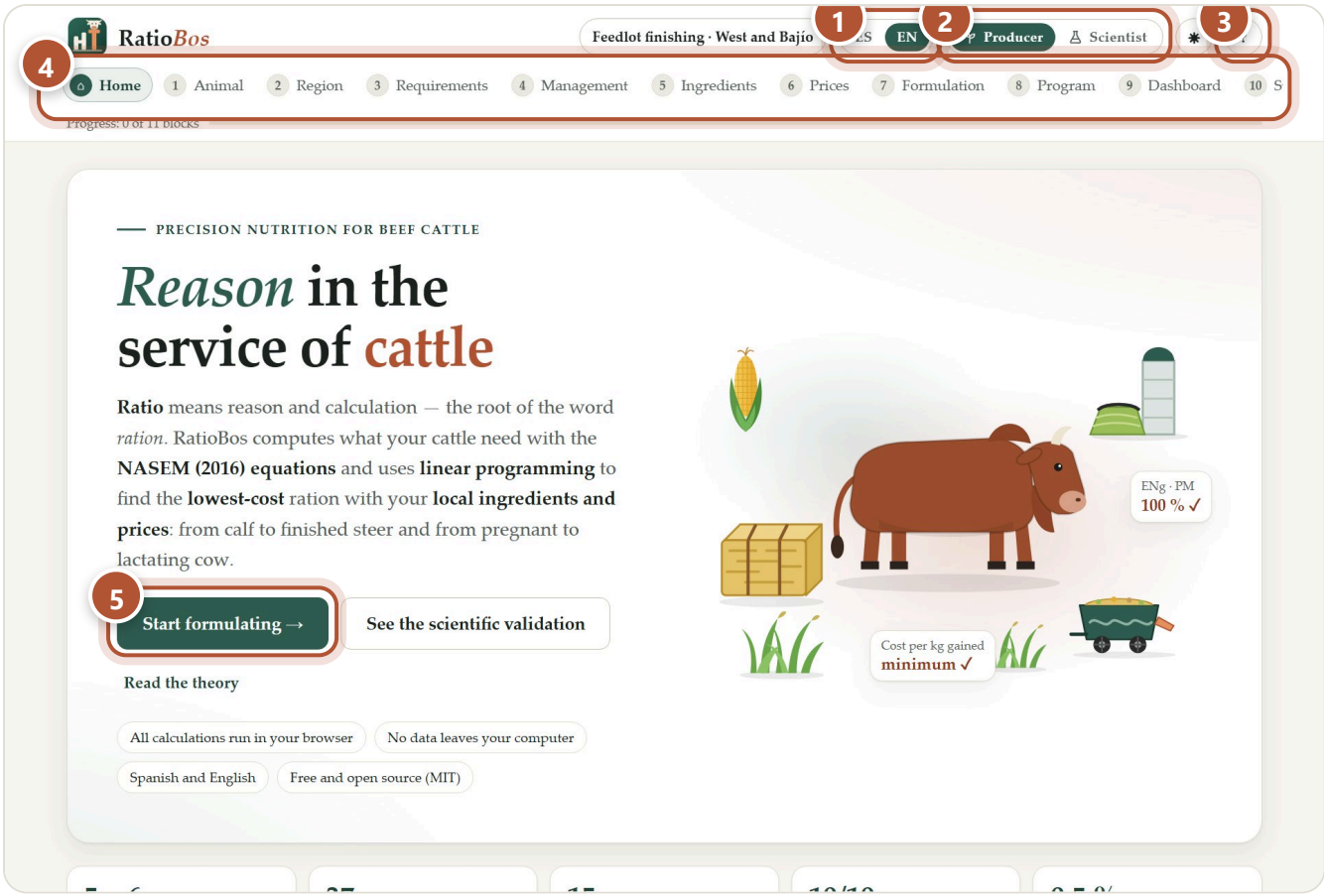



**FIGURE 0.1** **Home.** Everything you need to get around is in the top bar.

1. **ES / EN** switches the language of the whole app, reports included.
2. **Producer / Scientist** switches the view; your data is kept.
3. **?** opens the general help. Every field also has its own **?**.
4. The **block bar**, 1 to 11. You can jump to any of them; a tick marks the ones you have visited.
5. **Start formulating →** takes you to Block 1.

## 0.3 Two views, one app

The producer view shows only what you need to decide: the 8 key nutrients, a rumen-health traffic light, the mixer sheet in wet kilos, the expected gain and the cost per kilo produced. The scientist view adds the full model. Switch at any time, for instance to show your advisor where a number comes from.

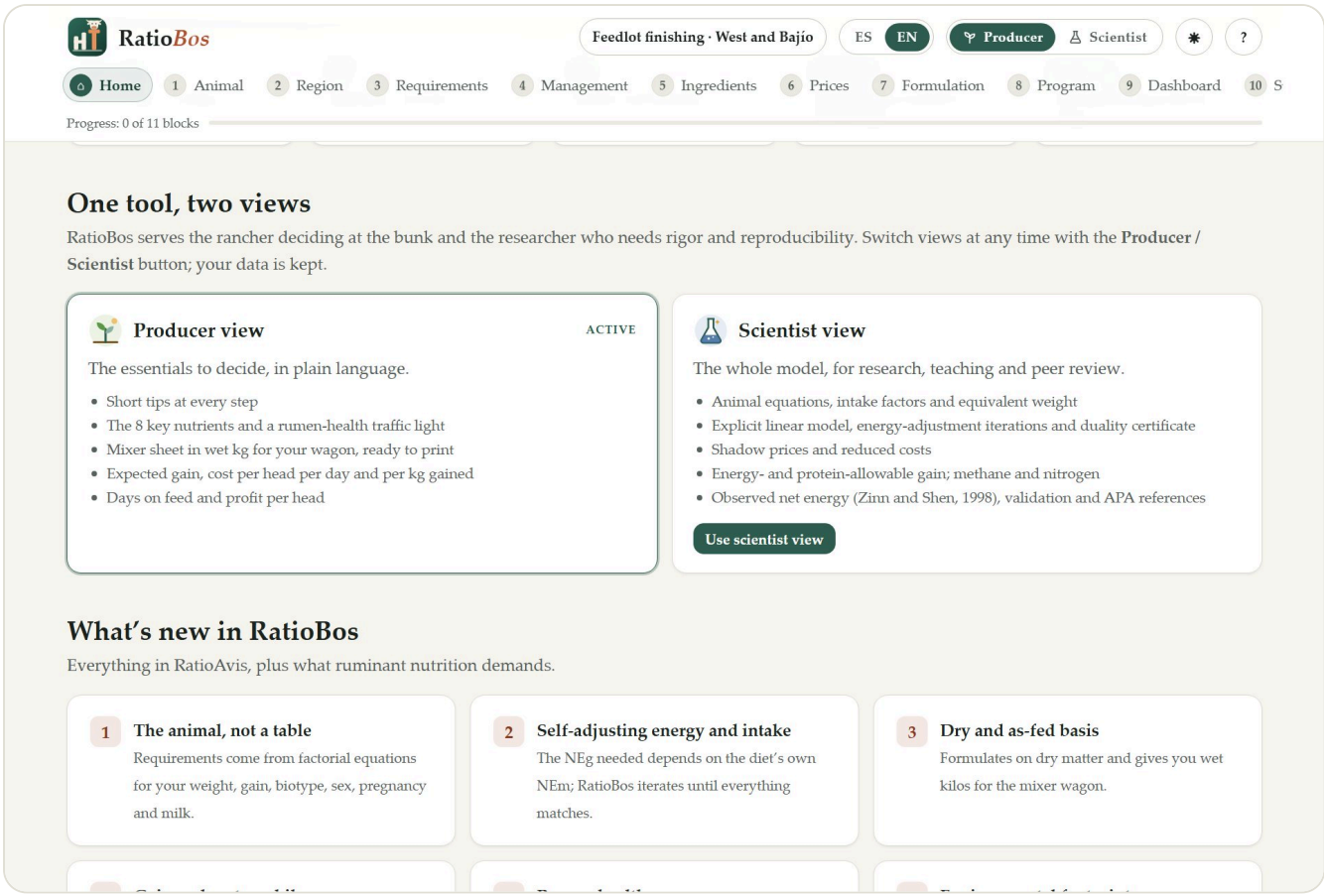



**FIGURE 0.2** **One tool, two views.** The active card has a border.

## 0.4 Help for every field

Hover (or tap, on a phone) over any **?**. You get what the field means and a colour-coded **decision scale**: green, fine; amber, watch; red, fix. It is the quickest way to learn without leaving the app.

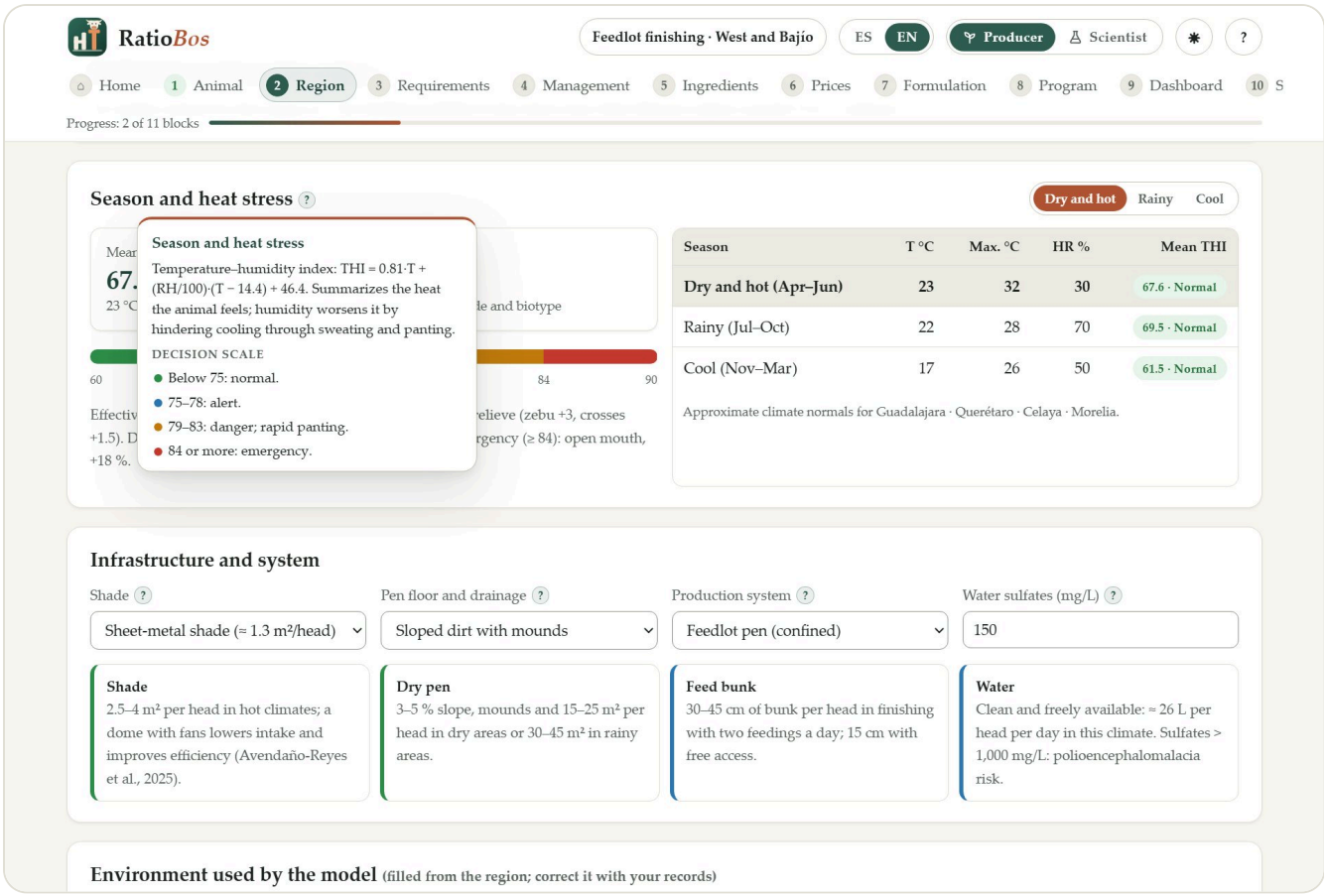



**FIGURE 0.3** **Help for the temperature-humidity index (THI).** Each level of the scale tells you what to do.

## 0.5 The case in this guide

So that every step has concrete numbers, we will follow one pen from start to finish:

| | | |
|---|---|---|
| ANIMALS<br>**300 intact bulls** | BIOTYPE<br>**Zebu × European cross** | ENTRY WEIGHT<br>**380 kg** |
| LOCATION<br>**Jalisco · West and Bajío** | SEASON<br>**Dry and hot (Apr–Jun)** | MIXER WAGON<br>**3,000 kg load** |

We buy at **$62** per kilo live weight and sell at **$58**. The target is **1.35 kg a day** up to 520 kg. It is a typical Bajío feedlot: local sorghum and corn, corn stover for fibre and a sheet-metal shade over the pens.

? **WHY SELL FOR LESS THAN YOU BUY?**

A light calf costs more per kilo than a finished steer. A feedlot makes money because each kilo the animal puts on costs less in feed than it is worth at sale. That is why the number that matters most in RatioBos is the **feed cost per kilo gained**.

## 0.6 The path in eleven steps

The first time takes about 25 minutes. After that, with your data saved, updating prices and reformulating takes under two.

**STEP 1** — 1 min
### Animal
Type, biotype and sex

**STEP 2** — 2 min
### Region
State, season and pens

**STEP 3** — 2 min
### Requirements
Weight and gain target

**STEP 4** — 2 min
### Management
Implant, lot and prices

**STEP 5** — 3 min
### Ingredients
The ones you can get

**STEP 6** — 5 min
### Prices
Delivered to the ranch

**STEP 7** — 1 min
### Formulation
The cheapest ration

**STEP 8** — 2 min
### Program
Days, cost and profit

**STEP 9** — 3 min
### Dashboard
Indicators and charts

**STEP 10** — 3 min
### Sustainability
Water, land and gases

**STEP 11** — 1 min
### Report
For your advisor

## 0.7 Your work saves itself

Every change is stored in that computer's browser. Close the tab and come back tomorrow, and everything is as you left it. To move it to another computer or back it up, use the path above (chapter 11).

BLOCK 1 · ANIMAL

# 1 Animal type, biotype and sex

First, know whom you are feeding. A calf, a finishing bull and a nursing cow need very different rations.

## 1.1 Choose the animal type

There are five cards. The first three use the **growth model** (the animal must gain weight); the last two, the **cow model** (maintain, carry a calf or make milk).

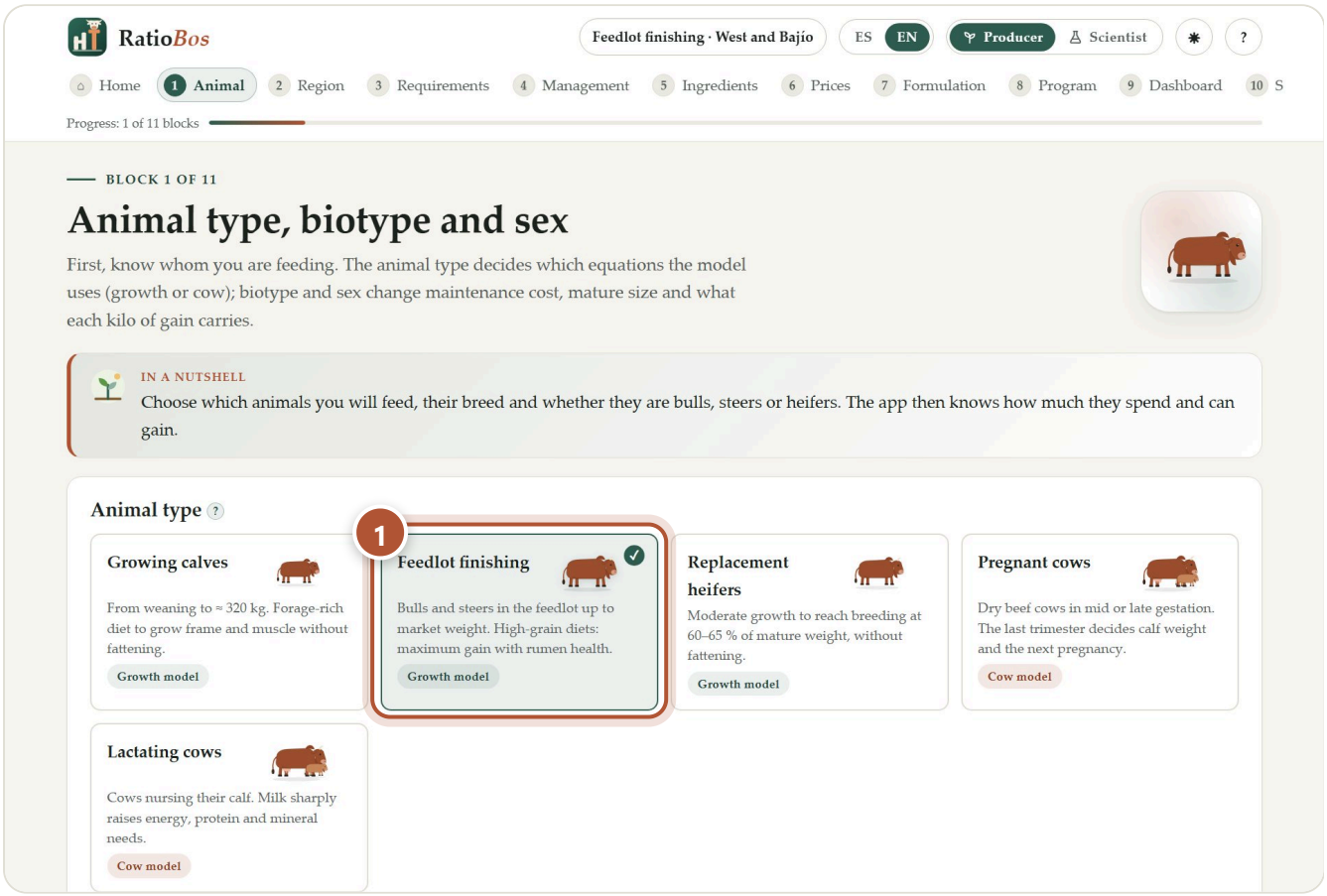



**FIGURE 1.1** **Block 1.** The selected card shows a tick and a label with the model it uses.

1 **Feedlot (finishing)**, the one in our case.

| TYPE | WHAT IT IS FOR |
|---|---|
| **Growing calves** | Weaning to ≈ 320 kg. Plenty of forage: build bone and muscle without fattening. |
| **Feedlot** | Bulls and steers up to sale weight. High-grain diets: top gain with a healthy rumen. |
| **Replacement heifers** | Moderate growth to reach breeding at 60–65 % of mature weight, without getting fat. |
| **Pregnant cows** | Dry cows in mid or late pregnancy. The last third decides calf weight and the next pregnancy. |
| **Lactating cows** | Nursing cows. Milk sharply raises energy, protein and mineral needs. |

## 1.2 Biotype and sex

The biotype groups breeds that behave alike in the pen. It changes how much the animal spends just staying alive (its **maintenance**) and at what weight it fattens. Sex does the same: an intact bull spends 15 % more on maintenance, but builds more muscle.

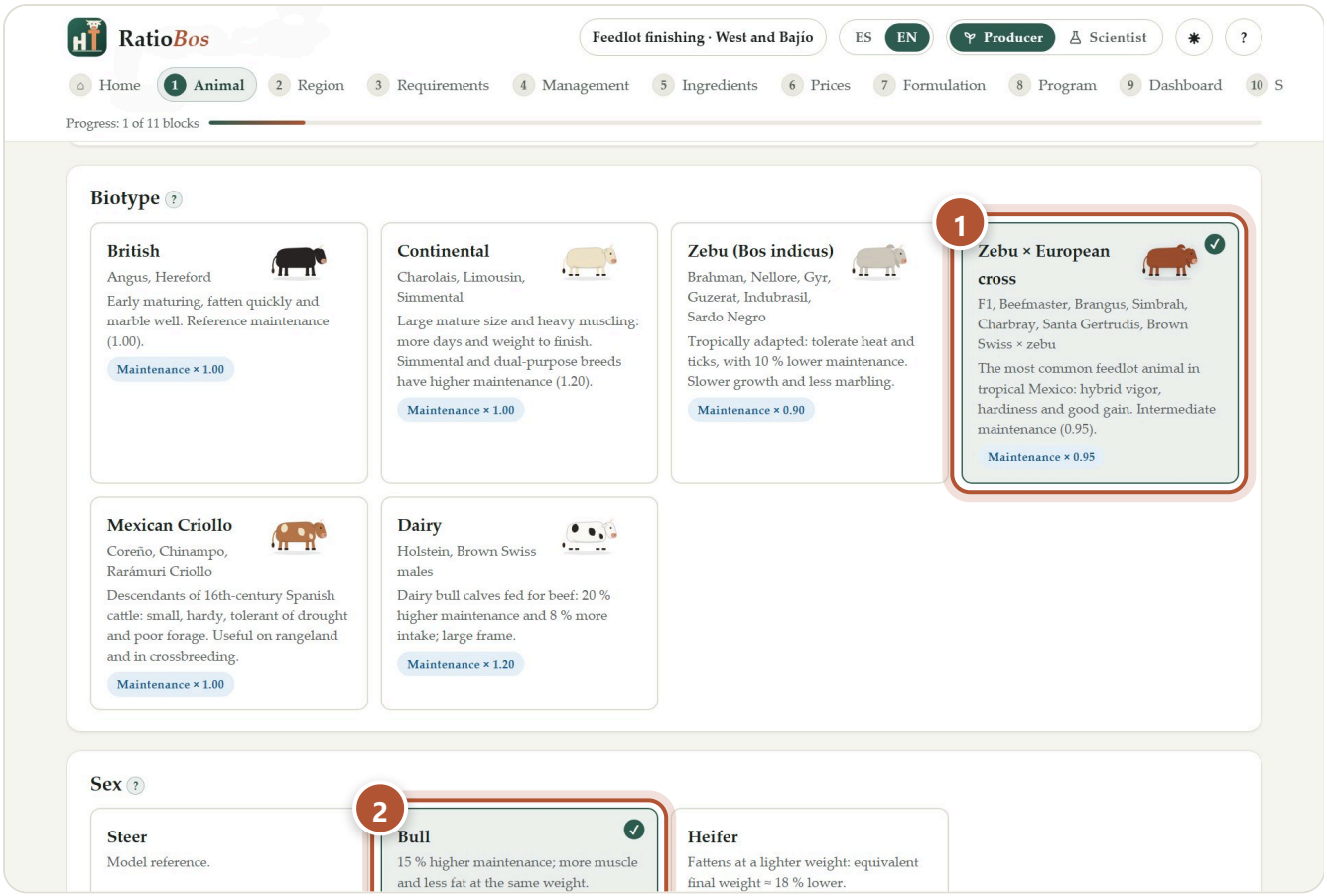



**FIGURE 1.2** **Biotypes and sex.** The blue label on each card is its maintenance factor.

1. **Zebu × European cross** (F1, Beefmaster, Brangus, Simbrah…): maintenance × 0.95.
2. **Bull (intact)**.

→ **DECISION RULE · WHICH BIOTYPE?**

| | |
|---|---|
| Angus, Hereford | **British** (× 1.00). Fatten early: sell lighter. |
| Charolais, Limousin, Simmental | **Continental**. More days and weight to finish. |
| Brahman, Nelore, Gyr, Indubrasil | **Zebu** (× 0.90). Handles heat; grows slower. |
| F1, Beefmaster, Brangus, Brown Swiss × zebu | **Cross** (× 0.95). The most common in Mexican feedlots. |
| Coreño, Chinampo, Rodeo Criollo | **Mexican Criollo**. Small and hardy; suited to rangeland. |
| Holstein, Brown Swiss males | **Dairy** (× 1.20). Spends more and eats 8 % more. |

◆ **IN THE PEN**

Our bulls are zebu × European crosses from a lot on the Jalisco coast. We tap **Feedlot**, **Zebu × European cross** and **Bull (intact)**, and move on with Next →.

BLOCK 2 · REGION

# Region, climate and infrastructure

2

Fattening cattle in Sonora is not the same as in Querétaro or Veracruz. Climate, grain prices and pen condition change gain and cost.

## 2.1 Tap your state on the map

The map groups Mexico's 32 states into **9 cattle regions**. Tap your state and the app loads the region's climate, soils, breeds, production systems, inputs and challenges, and below it shows published results from that region when available.

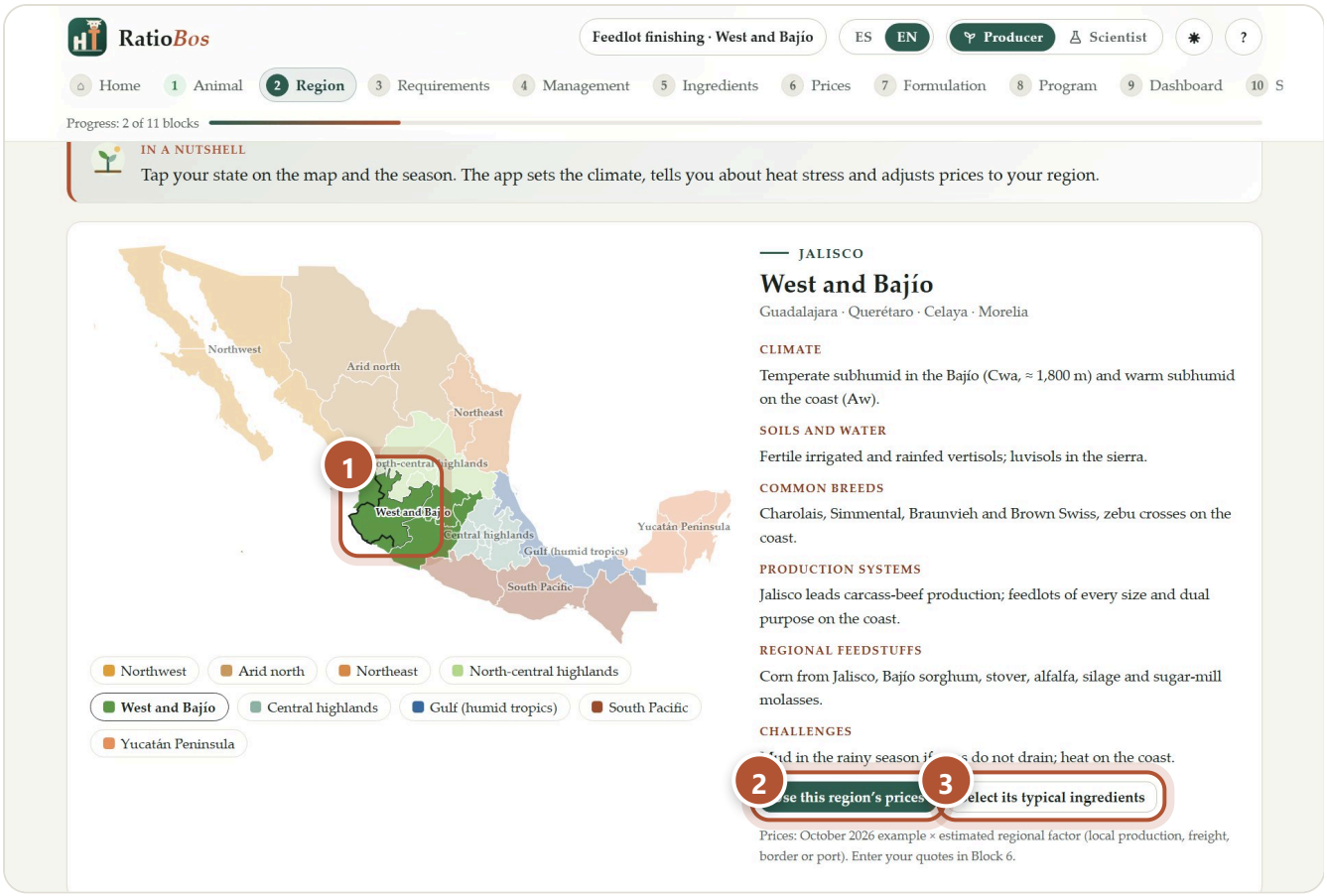



**FIGURE 2.1** **Block 2.** Jalisco, in the West and Bajío region.

1. The chosen state has a black outline; its region lights up.
2. **Use this region's prices** adjusts the sample prices by a regional factor (local supply, freight, border or port).
3. **Select its typical ingredients** ticks in Block 5 the feeds common in the region.

> **!** **CAUTION · REGIONAL PRICES ARE A GUIDE**
>
> They are for comparing regions and getting started. Before mixing, enter in Block 6 what **you** pay this week, delivered to the ranch.

## 2.2 Season and heat stress

Pick the season: **dry and hot** (April to June), **rainy** (July to October) or **cool** (November to March). The app computes the **temperature-humidity index (THI)**, which sums up the heat the animal feels: with high humidity it is harder to cool off by sweating and panting.

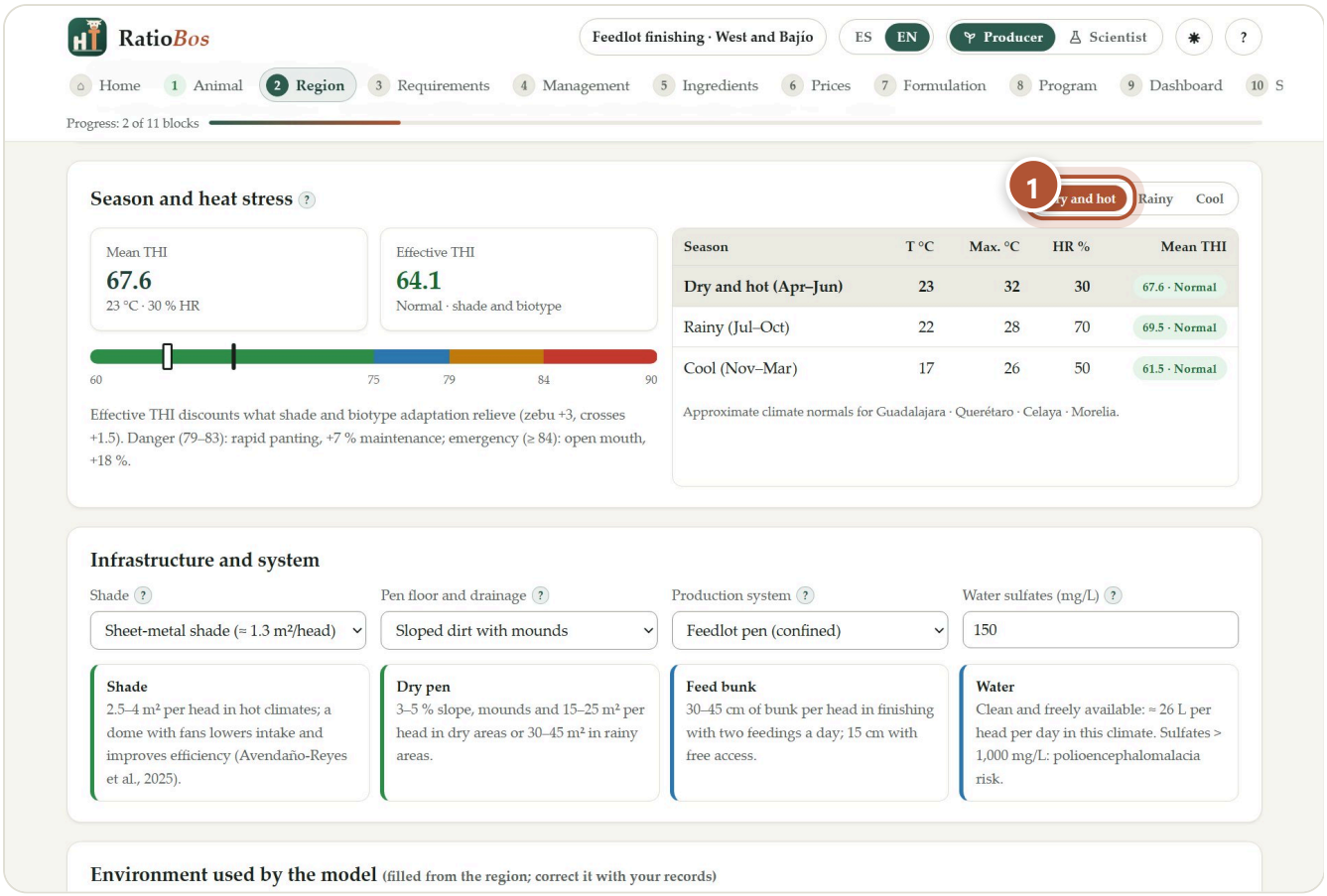



**FIGURE 2.2** **Season and THI.** The bar runs from green (normal) to red (emergency). The black tick is your effective THI.

1. The active season, in terracotta.

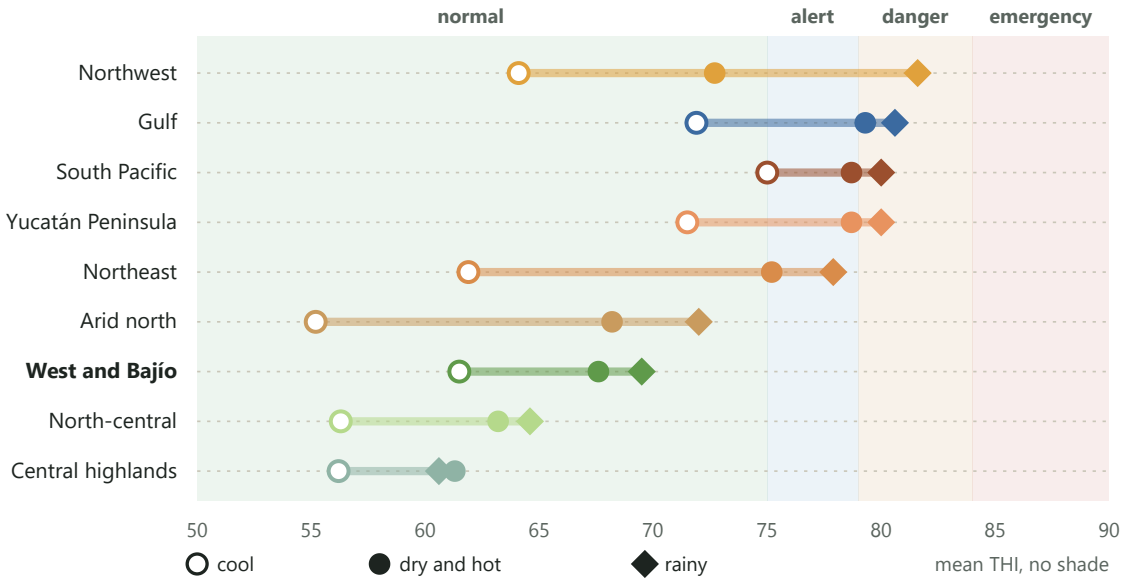



**FIGURE 2.3** **Mexico's heat by region and season.** Mean THI for each region (before shade or biotype credit). The humid tropics and coasts cross into the alert zone in the dry and rainy seasons; the highlands hardly ever do. Data from the app.

→ DECISION RULE · EFFECTIVE THI

| | |
|---|---|
| Below 75 · normal | No adjustment. Clean water, free access. |
| 75–78 · alert | Check shade and troughs. Feed more in the late afternoon. |
| 79–83 · danger | Rapid panting: maintenance rises 7 % and gain drops. Add shade. |
| 84 or more · emergency | Open-mouth panting: +18 % maintenance. Sprinklers or fans; do not move cattle in the heat of the day. |

## 2.3 Infrastructure and system

Four facts about your pens enter the model. Each has a card underneath with the practical recommendation.

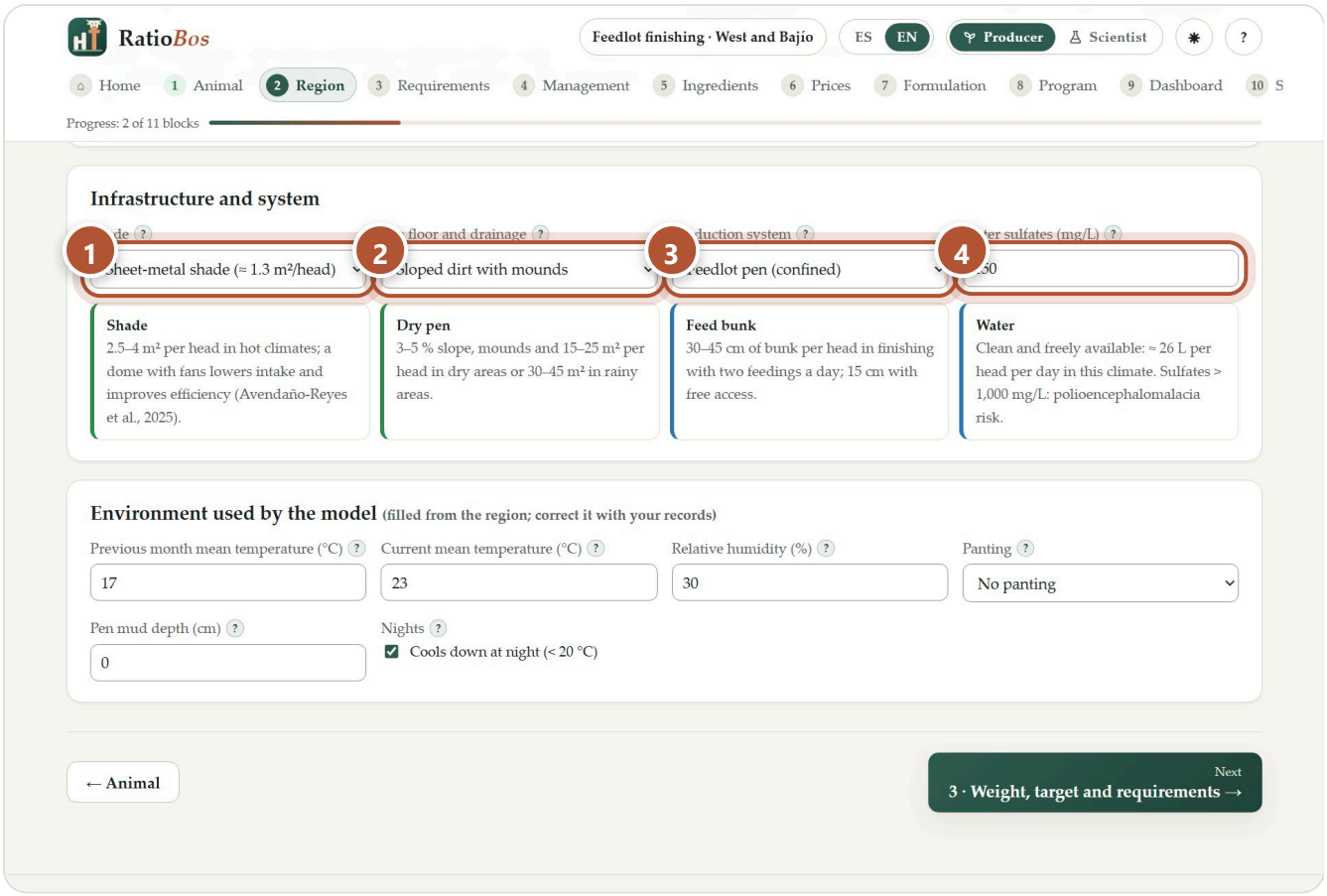



**FIGURE 2.4** **Infrastructure and system.**

1. **Shade:** takes points off the THI. A dome with fans helps more than a metal sheet.
2. **Floor and drainage:** decides how much mud there will be in the rains. Mud chills and tires the animal: it eats more and gains less.
3. **System:** on pasture the animal walks; on hills it spends up to 20 % more.
4. **Water sulfates:** water sulfur adds to the ration's. Above 1,000 mg/L there is a risk of polioencephalomalacia.

◆ **IN THE PEN**

We tap **Jalisco** and press **Use this region's prices**. In the **dry and hot** season the Bajío averages 23 °C and 30 % humidity: a THI of **67.6**, which drops to **64.1** with the metal shade and the cross's heat tolerance. Normal: heat does not hold our bulls back at this time of year. We keep a **sloped floor with mounds**, feedlot system and 150 mg/L sulfates.

BLOCK 3 · REQUIREMENTS

# 3 Weight, target and requirements

From your animals' weight and what you want to achieve, the app computes their daily needs with the NASEM (2016) equations.

## 3.1 Weight and gain target

Enter the lot's average **current live weight** and the **daily gain target**. If you weigh your cattle regularly and know what they really eat, enter the **actual DM intake**; leave it blank and the app predicts it. For cows, instead of gain you enter the days pregnant or the litres of milk.

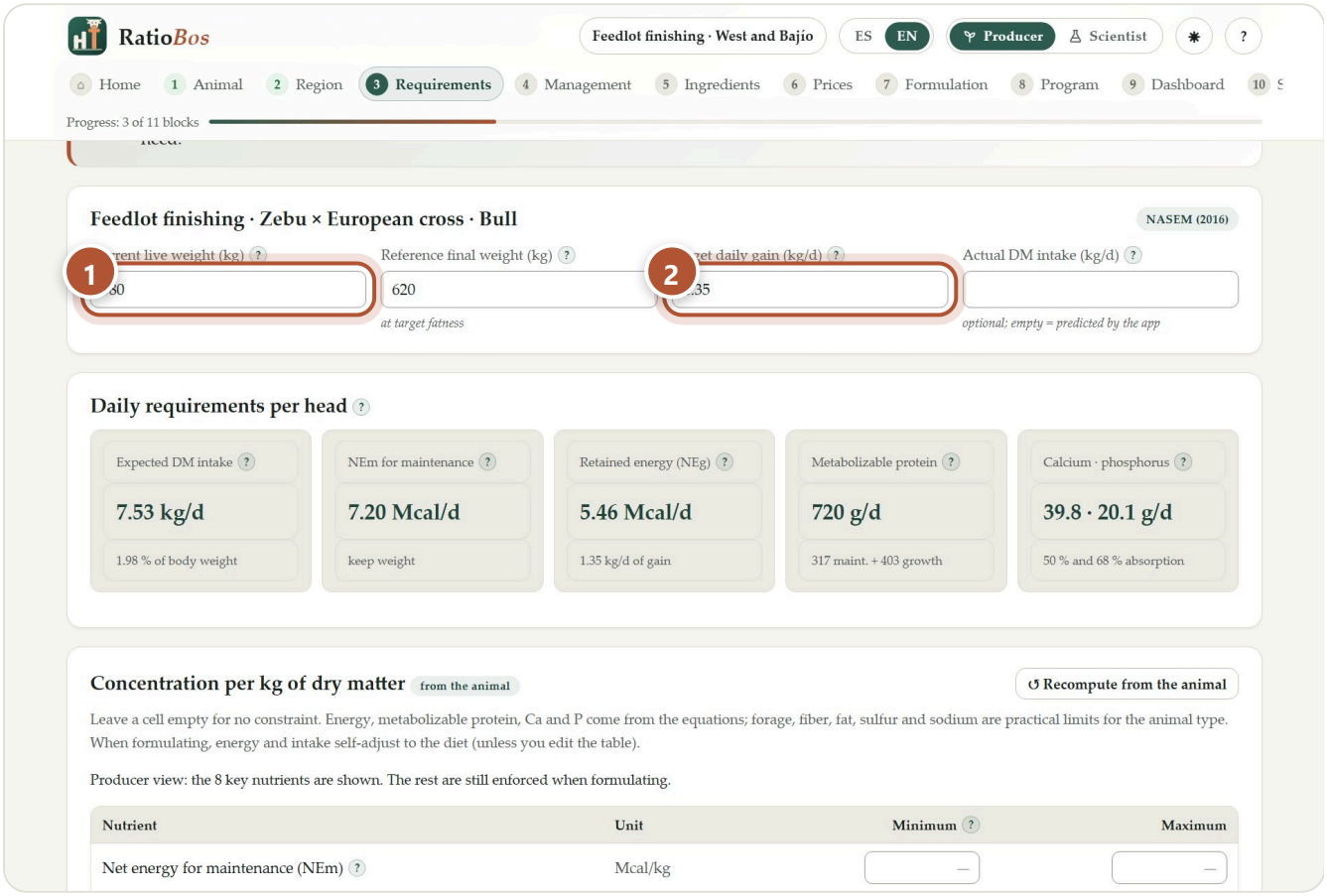



**FIGURE 3.1** **Block 3.** The animal's data.

1 **Current live weight:** 380 kg.

2 **Daily gain target:** 1.35 kg/d.

> ✓ **TIP · A REALISTIC TARGET**
>
> On high-grain feedlot diets, crosses gain 1.2–1.5 kg/d; zebu, 1.0–1.3; British and Continental, 1.4–1.7. Ask for more than the ingredients allow and the app will tell you the maximum feasible gain (chapter 7).

## 3.2 What each animal needs per day

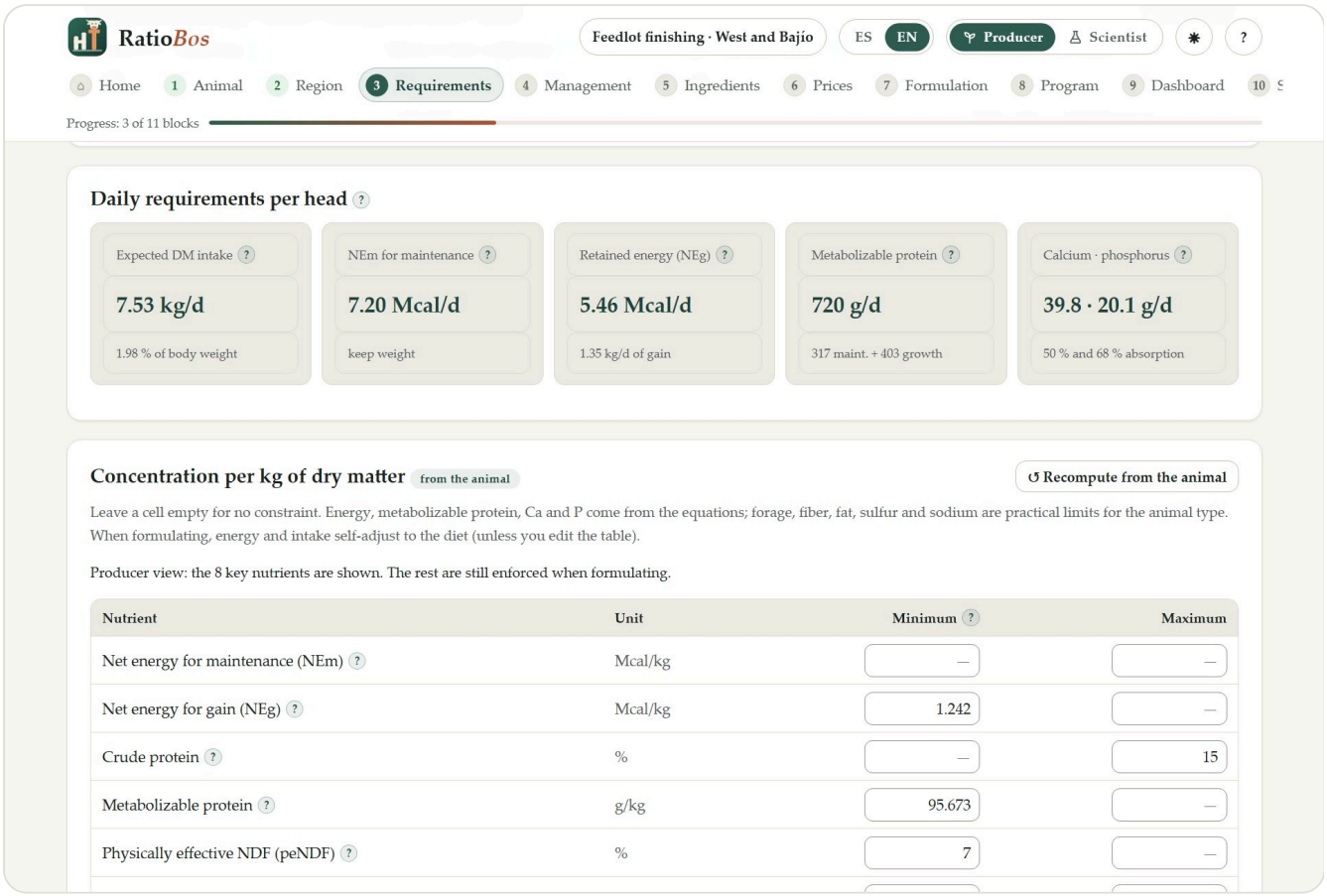



**FIGURE 3.2** **Daily requirements per animal** for the 380 kg bull that must gain 1.35 kg/d.

EXPECTED INTAKE
**7.53 kg**
DM per day · 1.98 % of weight

ENERGY TO MAINTAIN
**7.20 Mcal**
NEm per day

ENERGY TO GAIN
**5.46 Mcal**
NEg per day

METABOLIZABLE PROTEIN
**720 g**
317 maint. + 403 growth

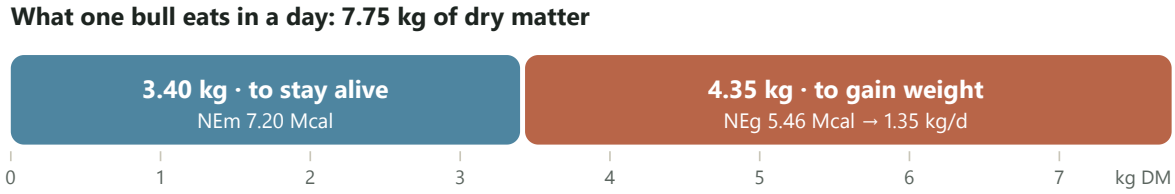



**FIGURE 3.3** **Where the feed goes.** Split of the case bull's intake between maintenance and gain, as computed by the app (section 7.5).

? **WHY TWO ENERGIES?**

Cattle use feed energy with different efficiency to maintain themselves than to gain weight. That is why the California system, used by NASEM, separates **net energy for maintenance (NEm)** from **net energy for gain (NEg)**. Maintenance is covered first; whatever is left becomes kilos.

## 3.3 The concentration table

The daily amounts become **how much each kilo of dry matter must contain**. Energy, metabolizable protein, calcium and phosphorus come from the equations; forage, effective fibre, fat, sulfur and sodium are practical limits for the animal type. The producer view shows the 8 key nutrients; the rest are still enforced when formulating.

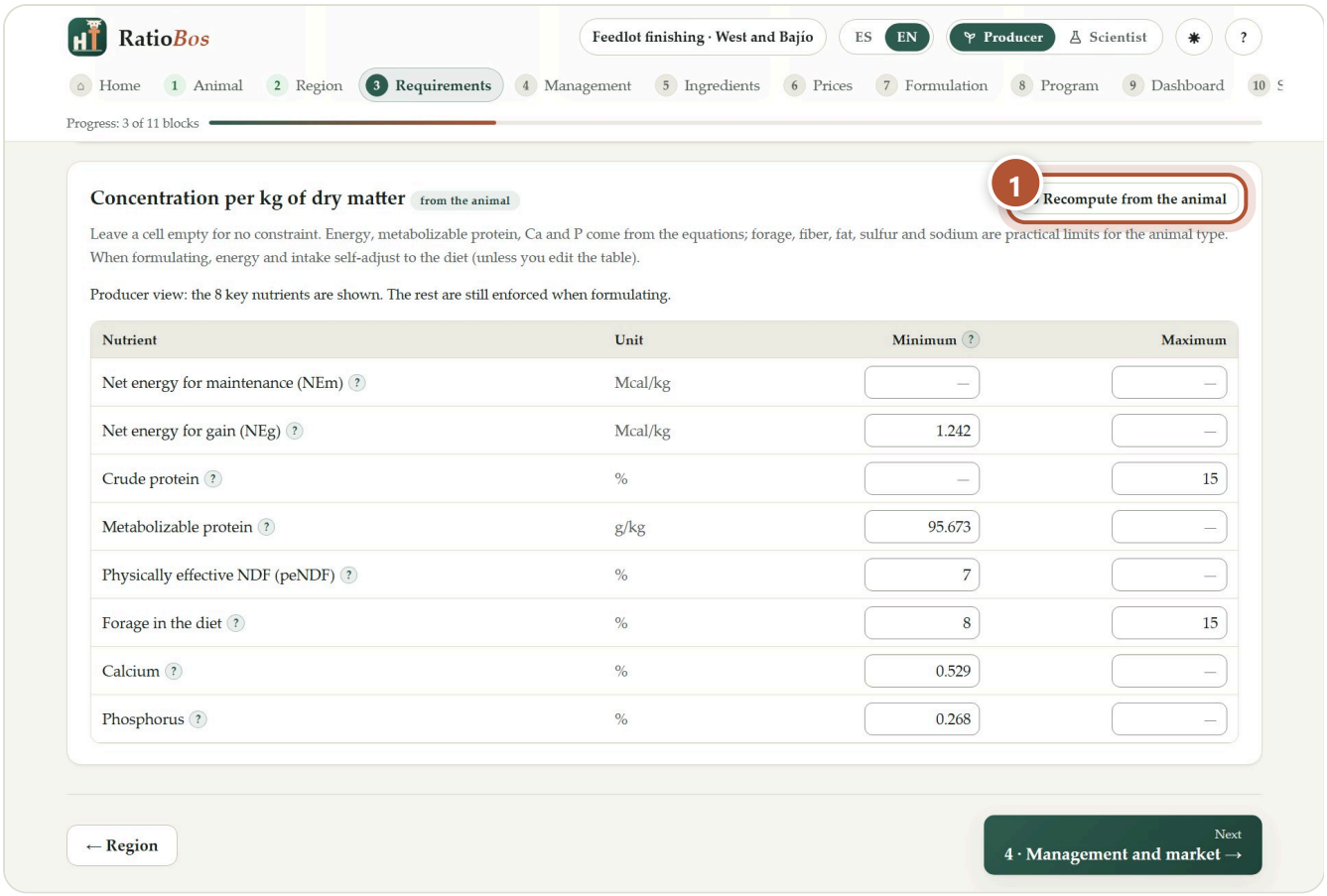



**FIGURE 3.4** **Concentration per kg of dry matter.** An empty cell does not constrain.

1 **↺ Recompute from the animal** goes back to the equation values if you edited anything.

> **! CAUTION · EDITING THE TABLE SWITCHES OFF THE AUTOMATIC ADJUSTMENT**
>
> The energy needed per kilo depends on the ration itself: a denser ration is eaten in smaller amounts. RatioBos adjusts energy and intake on its own until they match. If you type your own values, the app keeps yours and stops adjusting. Only do it if your nutritionist gave you a specification.

BLOCK 4 · MANAGEMENT

# 4 Management and market

Implants, ionophores and lot size change what the animal eats and yields; buying and selling prices decide whether the business makes a profit.

## 4.1 Management and health

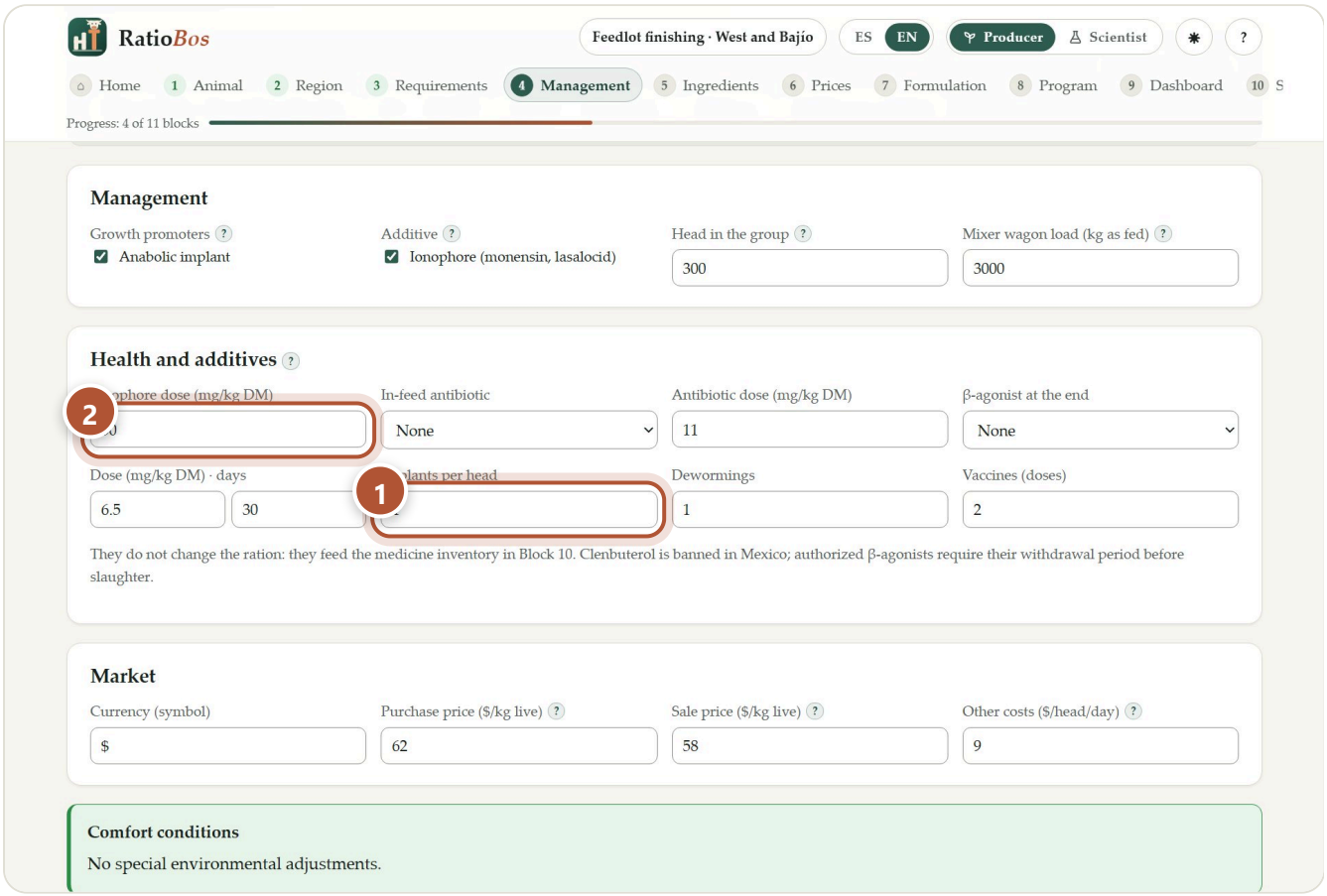



FIGURE 4.1 **Block 4 · Management.**

1. **Anabolic implant.** Without one, expected intake drops ≈ 6 %.
2. **Ionophore** (monensin, lasalocid). Lowers intake ≈ 3 % and improves energy use: the ration's NEm is multiplied by 1.12.

Also enter how many **animals are in the lot** and the **mixer-wagon load** in as-fed kilos: from those the app works out how many kilos to mix per day. The **Health and additives** card (ionophore dose, antibiotic, β-agonist, implants, dewormings and vaccines) does not change the ration: it feeds the medicine inventory in Block 10.

> **CAUTION · CLENBUTEROL IS BANNED**
>
> In Mexico clenbuterol is banned in livestock. Approved β-agonists (zilpaterol, ractopamine) require the withdrawal period before slaughter, and some export markets do not accept them.

## 4.2 Market

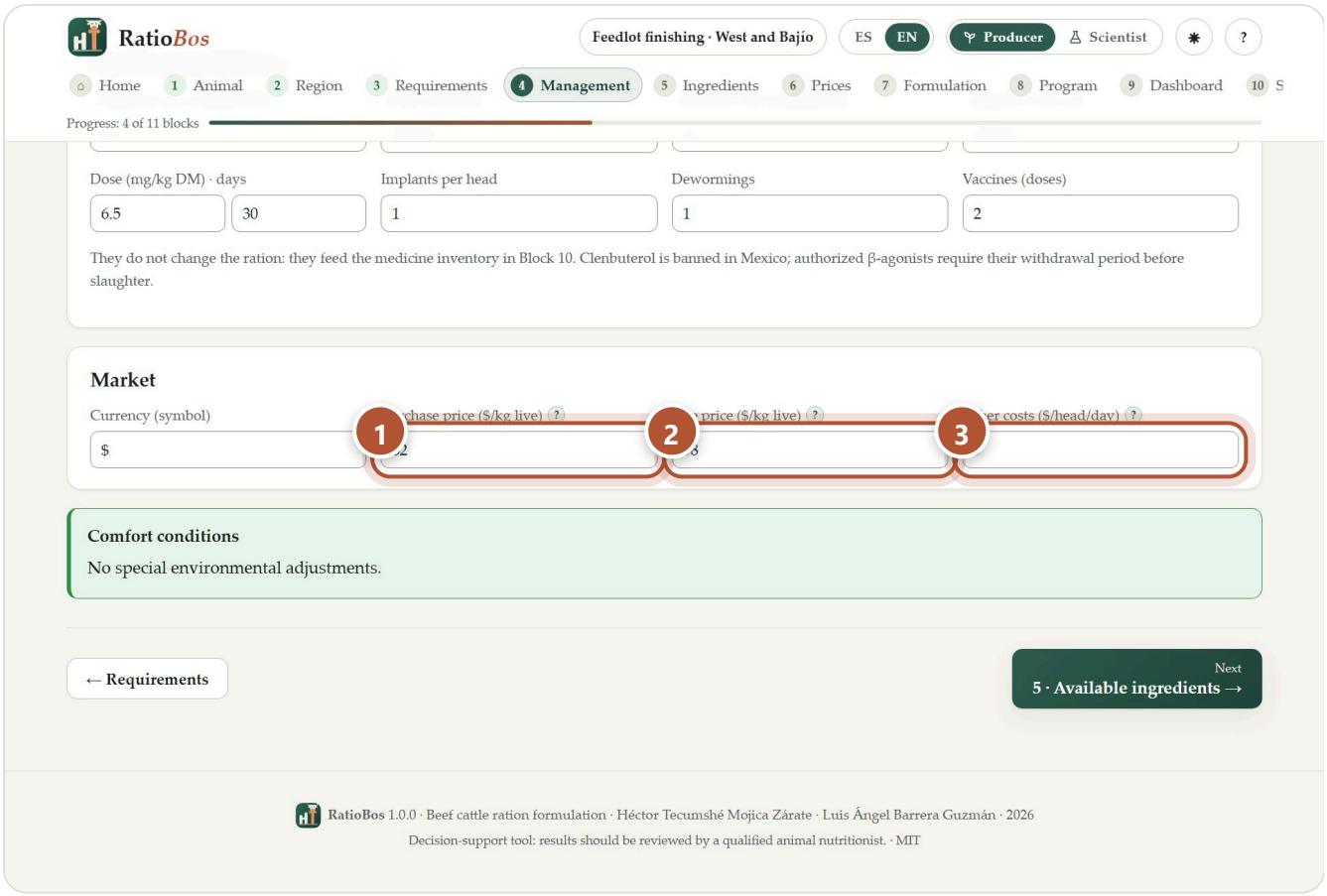



**FIGURE 4.2** **Market.** Prices per kilo live weight.

1. **Purchase price** of the animal entering the pen.
2. **Sale price** of the finished animal.
3. **Other costs** per head per day: labour, health, power, interest on capital.

> **IN THE PEN**
>
> We keep implant and ionophore ticked (30 mg monensin per kg DM), type `300` animals and a `3000` kg load. In the market card: buy `62`, sell `58` and `9` pesos of other costs per head per day.

BLOCK 5 · INGREDIENTS

# 5 Available ingredients

Tick only what you can actually get. The more options the app has, the cheaper the ration can be.

## 5.1 The ingredient library

RatioBos ships with **37 ingredients** grouped into dry forages, silages and fresh forages, grains, by-products, protein sources, non-protein nitrogen, fats, minerals and premixes. Their values come from the Oklahoma State University table (Beck, Lalman & Moehlenpah, 2024) with NASEM (2016) data, and all are on a **dry-matter basis**.

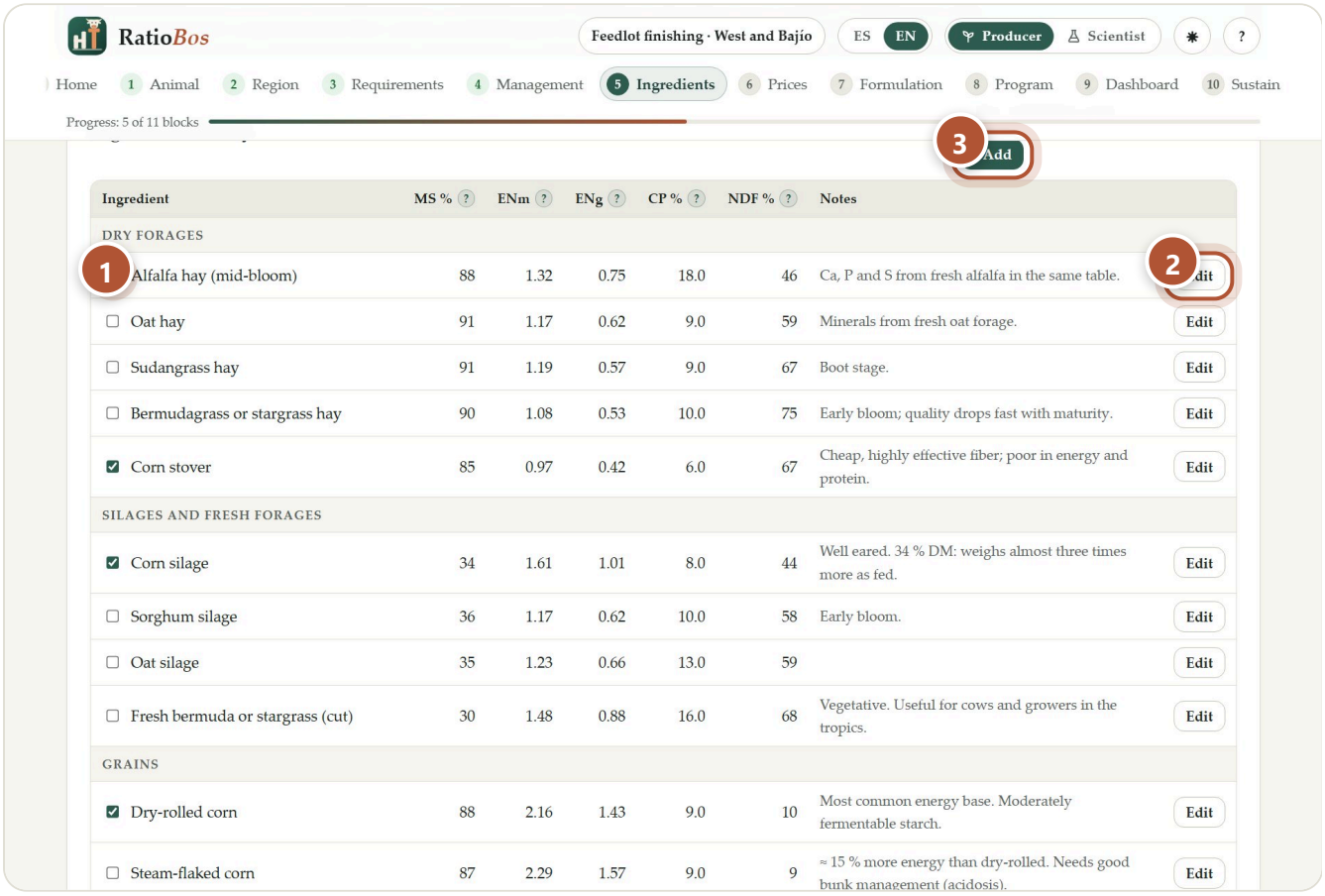



**FIGURE 5.1** **Block 5.** One row per ingredient, with dry matter, energy, protein and fibre.

1. The **checkbox** makes it available to the optimiser.
2. **Edit** opens its full composition.
3. **+ Add** creates a new ingredient (a local by-product, for instance).

> **?** **WHY ON A DRY BASIS?**
>
> Corn silage is 34 % dry matter: two thirds of its weight is water. Comparing a kilo of silage with a kilo of sorghum (90 % DM) would be unfair. On a dry basis everything is compared by what actually feeds; at the end the app gives you back the wet kilos for the wagon.

## 5.2 Use your lab analyses

If you sent your silage or stover for analysis, press **Edit** and change the values. What weighs most in the result is **dry matter**, **crude protein** and **NDF**; leave the energy if the lab does not report it.

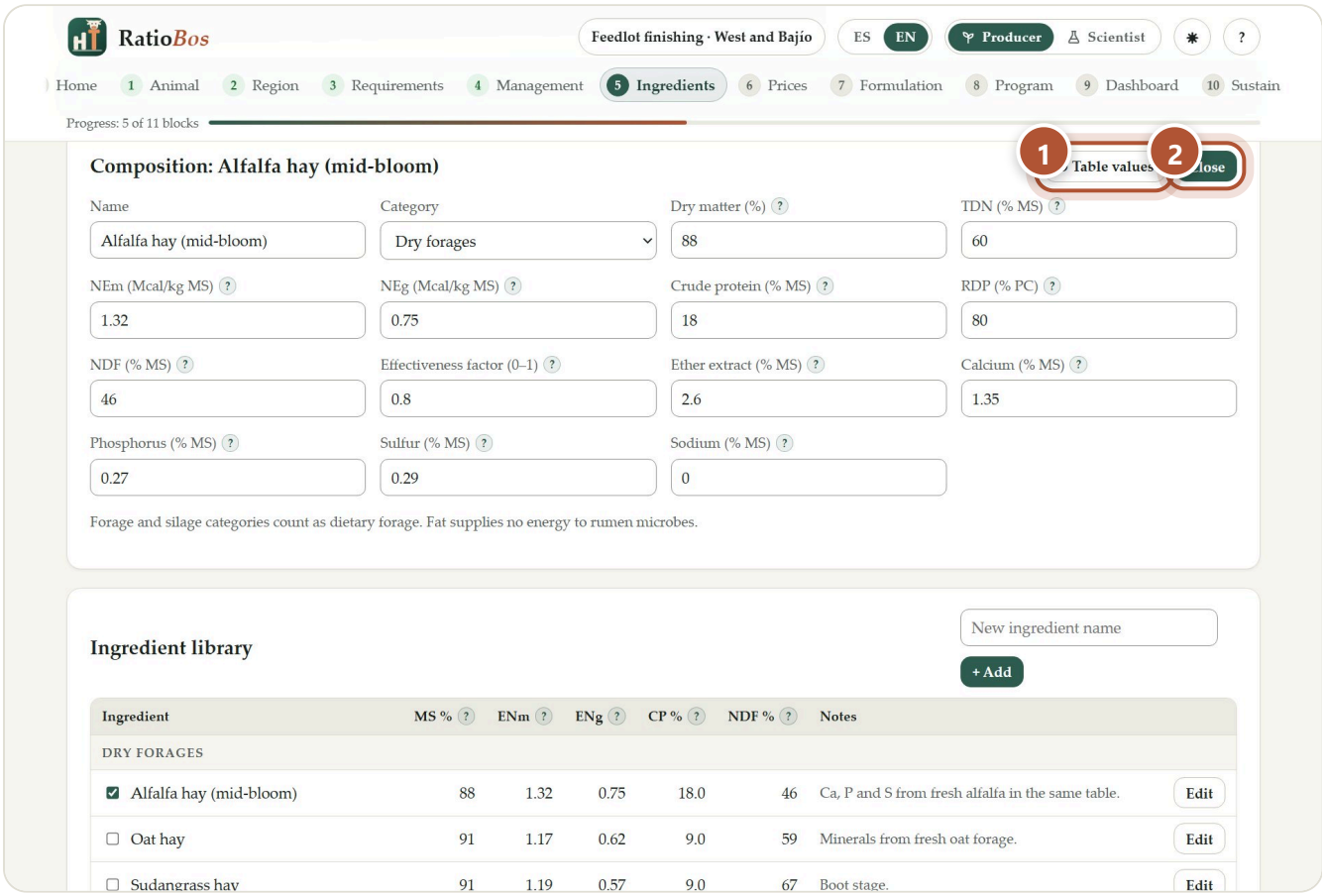



**FIGURE 5.2** **An ingredient's composition.** Every field has its **?**.

1 **↺ Table values** undoes your changes. 2 **Close**.

> **◆** **IN THE PEN**
>
> We have 12 ingredients ticked: alfalfa hay, corn stover, corn silage, dry-rolled corn, dry-rolled sorghum, molasses, DDGS, 44 % soybean meal, urea, limestone, salt and the trace-mineral and vitamin premix.

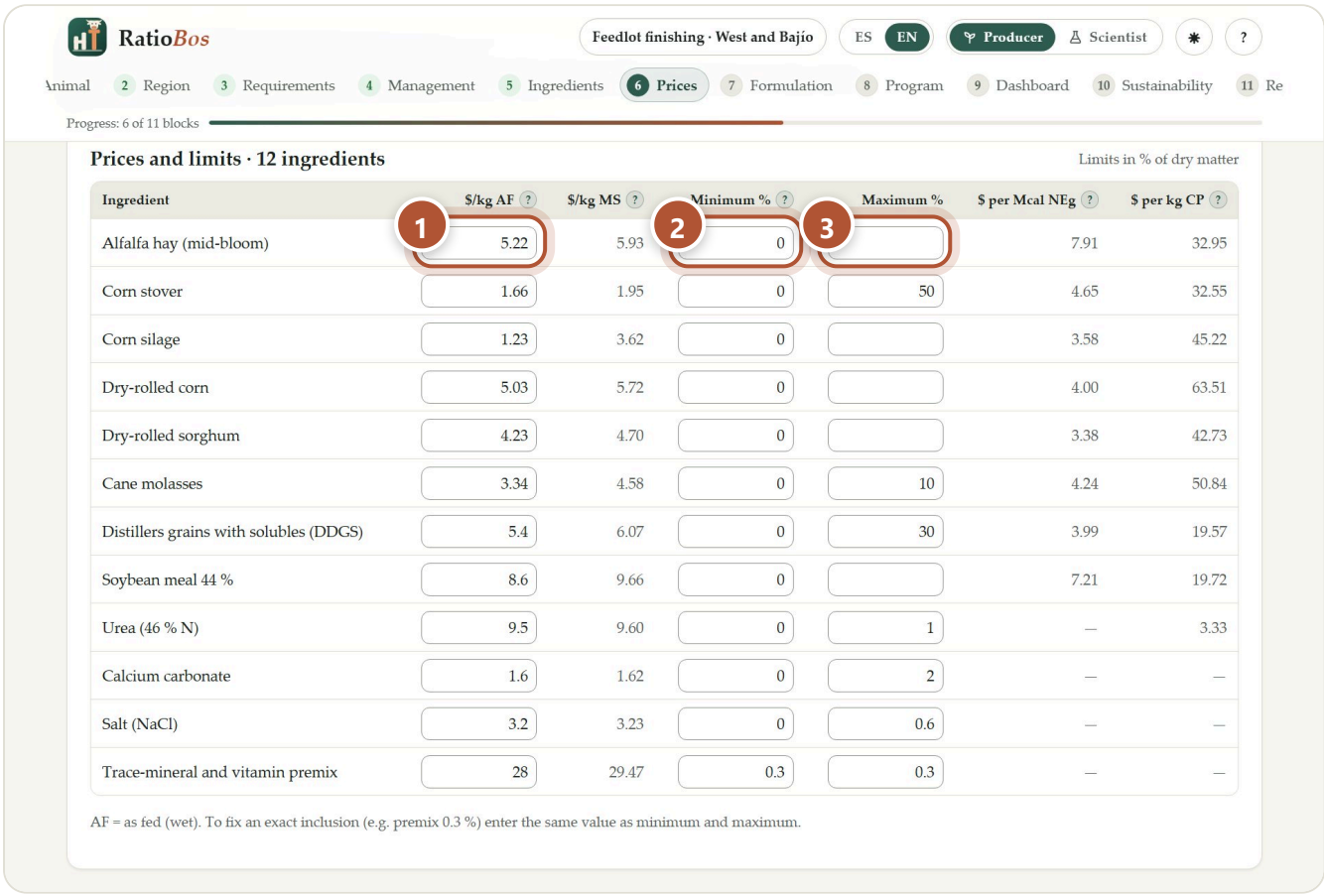
BLOCK 6 · PRICES

# 6 Local prices and inclusion limits

The price you enter here is what the app minimises. A stale price gives a ration that is no longer the cheapest.

## 6.1 Enter what you pay

Enter the price per kilo **as you buy it** and **delivered to the ranch** (freight and unloading included). The app converts it to price per kilo of dry matter, which is what it compares.




| Ingredient | $/kg AF | $/kg MS | Minimum % | Maximum % | $ per Mcal NEg | $ per kg CP |
|---|---|---|---|---|---|---|
| Alfalfa hay (mid-bloom) | 5.22 | 5.93 | 0 | | 7.91 | 32.95 |
| Corn stover | 1.66 | 1.95 | 0 | 50 | 4.65 | 32.55 |
| Corn silage | 1.23 | 3.62 | 0 | | 3.58 | 45.22 |
| Dry-rolled corn | 5.03 | 5.72 | 0 | | 4.00 | 63.51 |
| Dry-rolled sorghum | 4.23 | 4.70 | 0 | | 3.38 | 42.73 |
| Cane molasses | 3.34 | 4.58 | 0 | 10 | 4.24 | 50.84 |
| Distillers grains with solubles (DDGS) | 5.4 | 6.07 | 0 | 30 | 3.99 | 19.57 |
| Soybean meal 44 % | 8.6 | 9.66 | 0 | | 7.21 | 19.72 |
| Urea (46 % N) | 9.5 | 9.60 | 0 | 1 | – | 3.33 |
| Calcium carbonate | 1.6 | 1.62 | 0 | 2 | – | – |
| Salt (NaCl) | 3.2 | 3.23 | 0 | 0.6 | – | – |
| Trace-mineral and vitamin premix | 28 | 29.47 | 0.3 | 0.3 | – | – |

AF = as fed (wet). To fix an exact inclusion (e.g. premix 0.3 %) enter the same value as minimum and maximum.

FIGURE 6.1 **Block 6.** Prices and limits for the 12 ticked ingredients.

1. **$/kg AF**: your price, as fed.
2. **Minimum %** of dry matter: forces at least that much in.
3. **Maximum %**: the cap for animal health, palatability or logistics.

## 6.2 Compare prices with two columns

The **$ per Mcal NEg** and **$ per kg CP** columns tell you what each unit of energy and protein costs in each ingredient. That is the right way to compare: the sorghum in our case gives the cheapest energy ($3.38 per Mcal) and DDGS the cheapest protein after urea ($19.57 per kg CP).

→ **DECISION RULE · LIMITS WORTH KEEPING**

| | |
|---|---|
| **Urea** | At most 1 % of DM, always well mixed. More risks toxicity. |
| **DDGS** | At most 30 %: they bring sulfur and fat. |
| **Molasses** | At most 10 % in grain-based finishing. |
| **Whole cottonseed** | At most 15 % because of gossypol. |
| **Limestone** | At most 2 %: it is a calcium source, not a filler. |
| **Premix** | Minimum = maximum = the maker's dose (0.3 % in our case). |

✓ **TIP · TO FIX AN EXACT INCLUSION**

Type the same value as minimum and maximum. Use it for the premix or for an ingredient you have already bought and want to use up.

BLOCK 7 · FORMULATION

# 7 Least-cost formulation

One button. The app searches every possible blend for the cheapest one that covers what your cattle need, and tells you how much they will gain and what each kilo costs.

## 7.1 Press Formulate

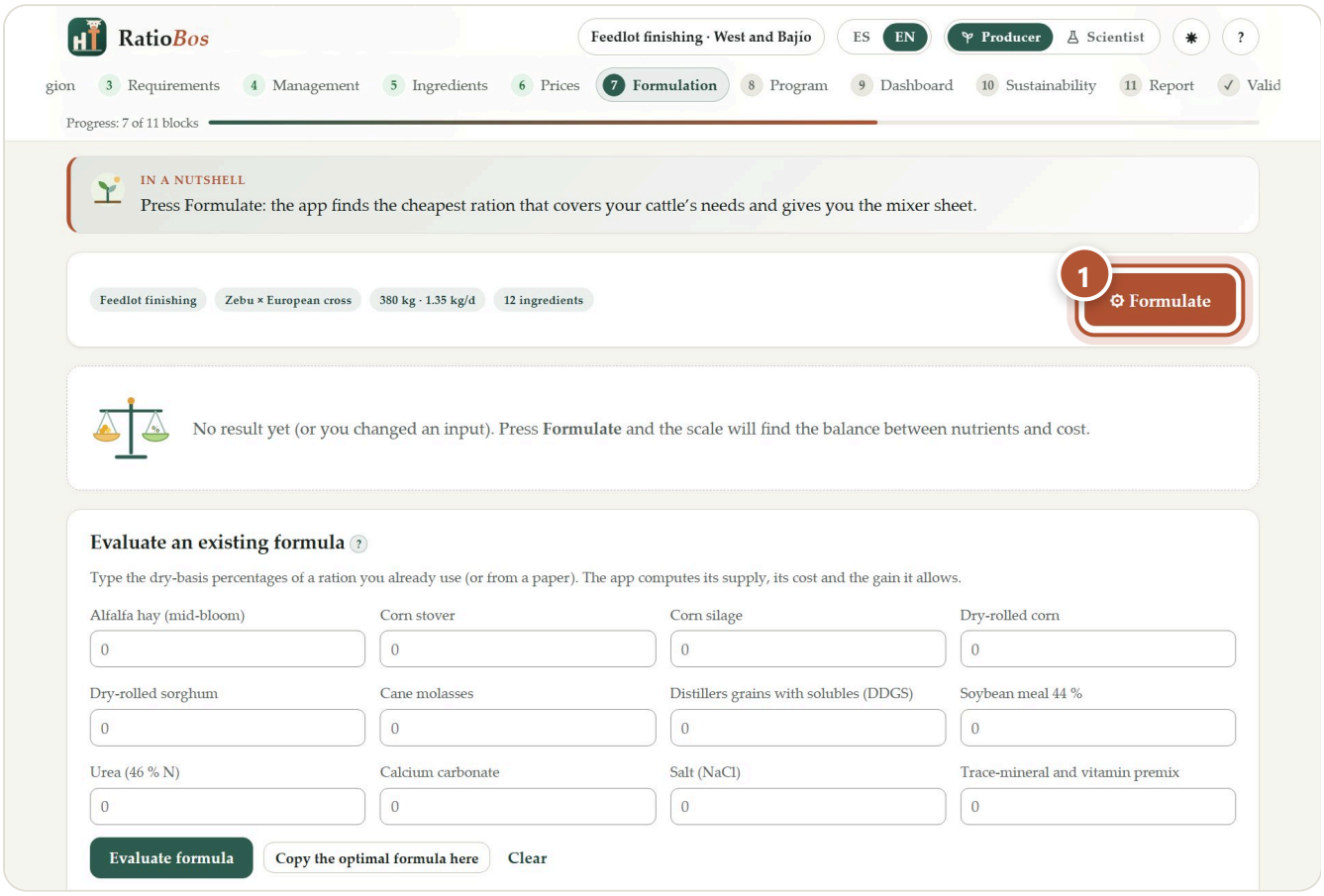



FIGURE 7.1 **Block 7.** At the top, a summary of what you are about to formulate.

1 **Formulate**. It takes under a second.

**?** **WHAT DOES THE APP DO IN THAT SECOND?**

It solves a **linear programming** problem: among every combination of your ingredients that meets each minimum and maximum, it finds the one with the lowest cost per kilo of dry matter. Then it checks whether that ration's energy matches the one it assumed to estimate intake; if not, it corrects and solves again until they match. Finally it proves with a mathematical certificate that no cheaper one exists.

## 7.2 The four result cards

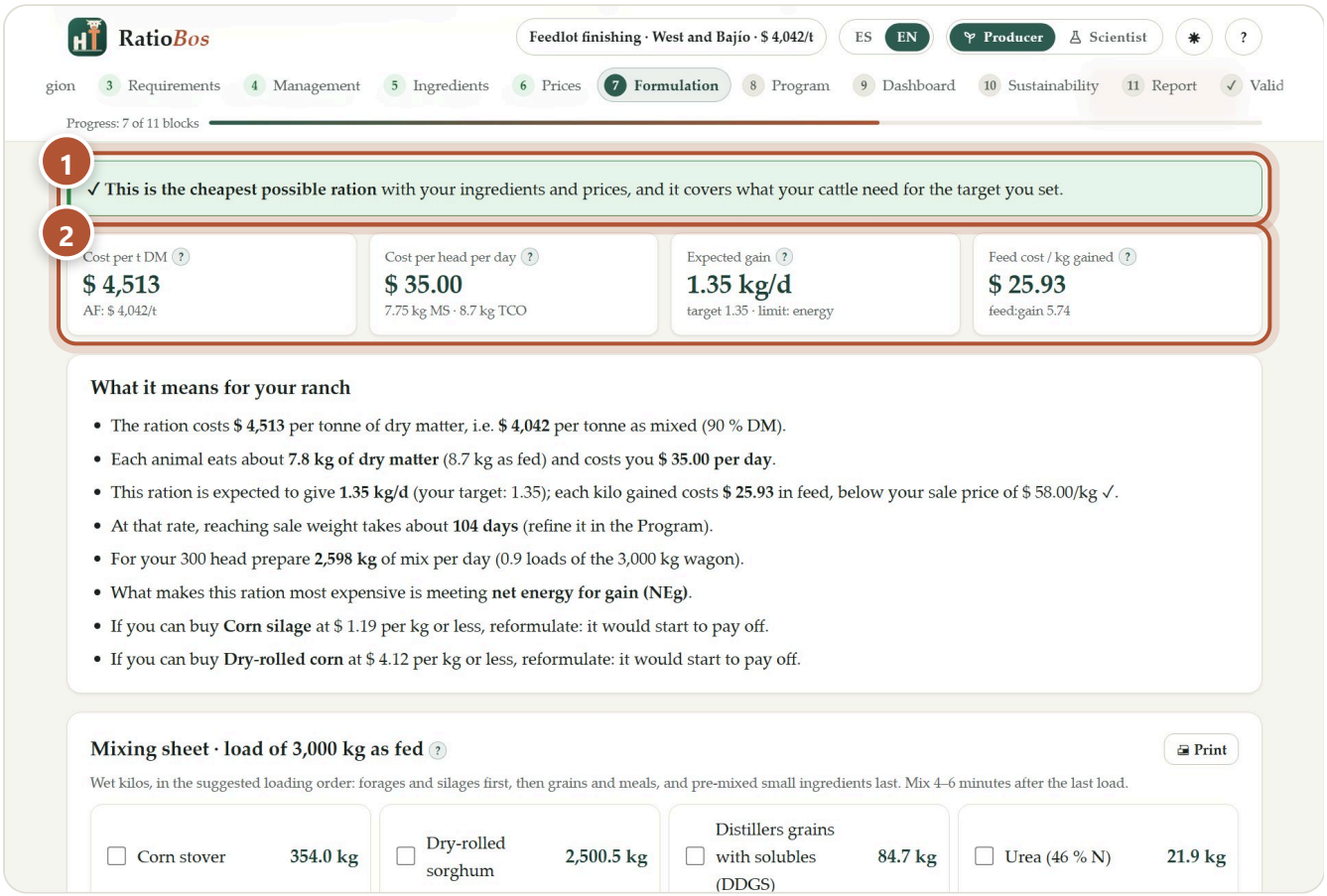



**FIGURE 7.2** **Result.** The green message confirms it is the cheapest possible ration.

1 Least-cost confirmation.

2 The **four cards** explained below.

COST PER T DM
**$ 4,513**
$ 4,042 per t as fed

PER HEAD PER DAY
**$ 35.00**
7.75 kg DM · 8.7 kg AF

EXPECTED GAIN
**1.35 kg/d**
limited by: energy

FEED PER KG GAINED
**$ 25.93**
feed conversion 5.74

| CARD | WHAT IT TELLS YOU |
|---|---|
| **Cost per t DM** | What a tonne of the ration costs without water. Below, the same tonne wet, as it leaves the wagon. |
| **Cost per head per day** | Expected intake × price. Each head's daily feed bill. |
| **Expected gain** | The gain the ration allows, and what limits it: energy or protein. |
| **Feed cost per kg gained** | The business number. It must sit well below your sale price. |

## 7.3 What it means for your ranch

Below the cards, the app turns the numbers into sentences: how much each animal eats, how many days remain to sale weight, how much to mix per day and which ingredient would start to pay off if its price fell.

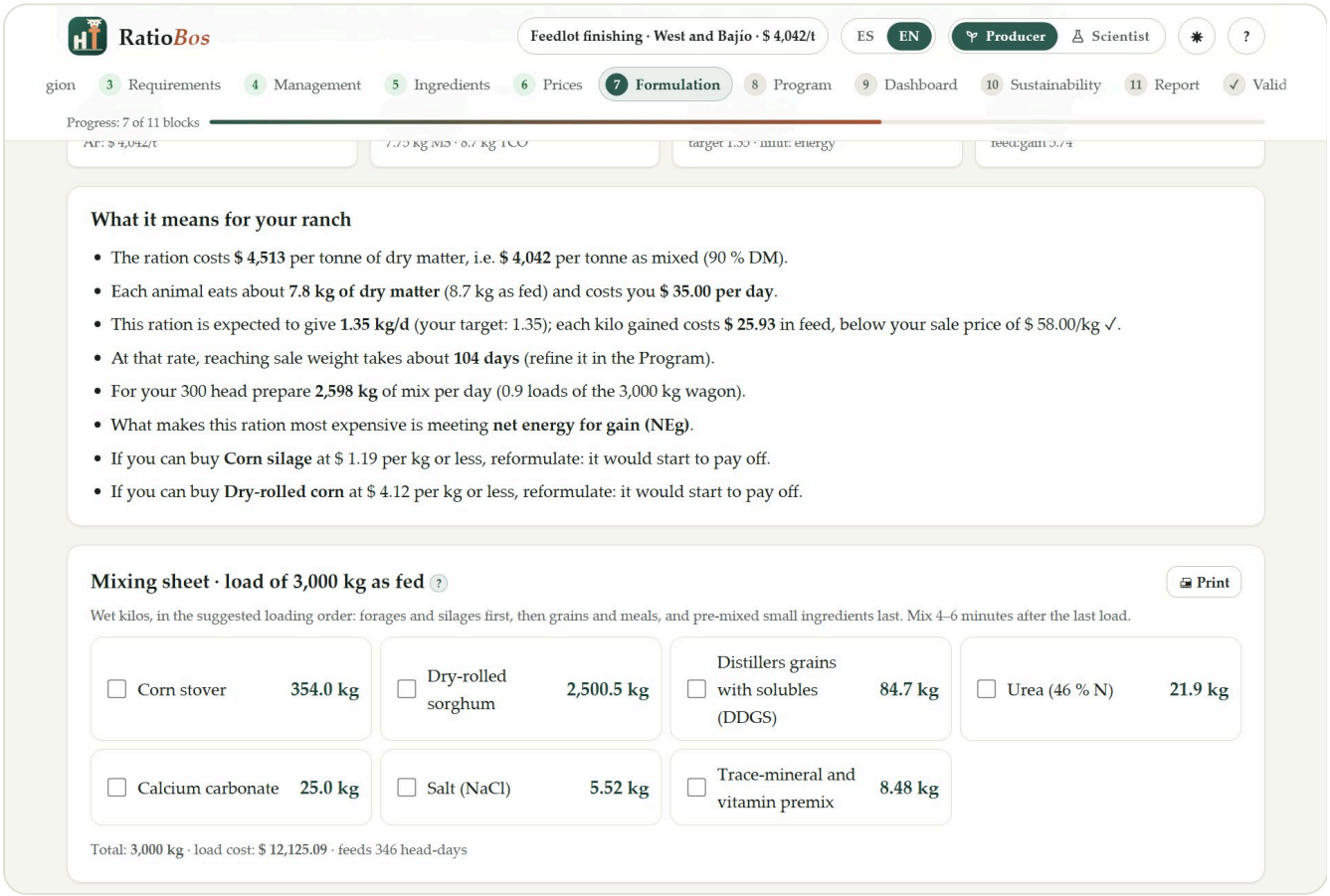



**FIGURE 7.3** **What it means for your ranch.** Always in plain language.

> **IN THE PEN**
>
> The ration costs **$4,042 per tonne** as mixed. Each bull eats 8.7 kg a day and costs **$35.00**. Expected gain is **1.35 kg/d**, at **$25.93 of feed per kilo gained**: less than half the $58 we will sell it for. For the 300 animals we must prepare **2,598 kg a day**, just under one wagon load, and they will reach sale weight in about 104 days.

## 7.4 The mixer-wagon sheet

This is what goes to the pen. The kilos are **wet**, as the wagon scale weighs them, and come in **loading order**: forages and silages first, then grains and meals, and the small pre-blended ingredients last. Each line has a box to tick as you load.

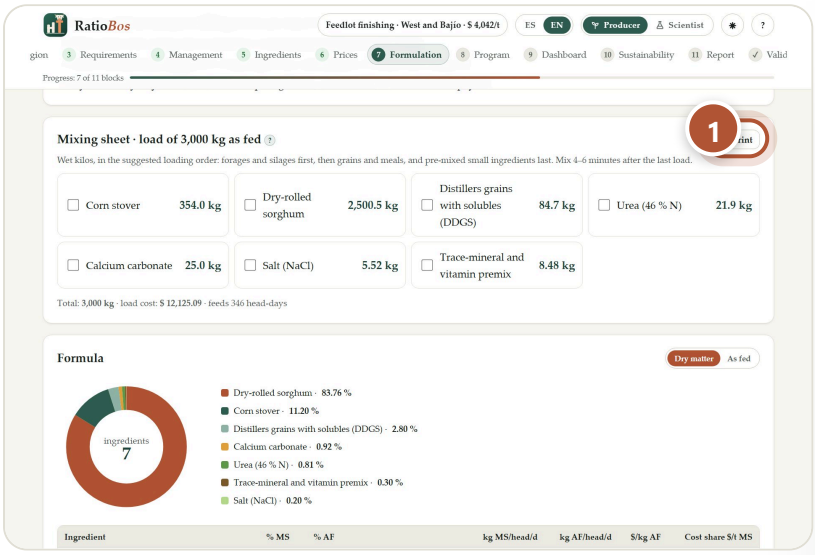



FIGURE 7.4 **Mixing sheet** for a 3,000 kg load.

1 **Print** prints just the sheet, on one page.

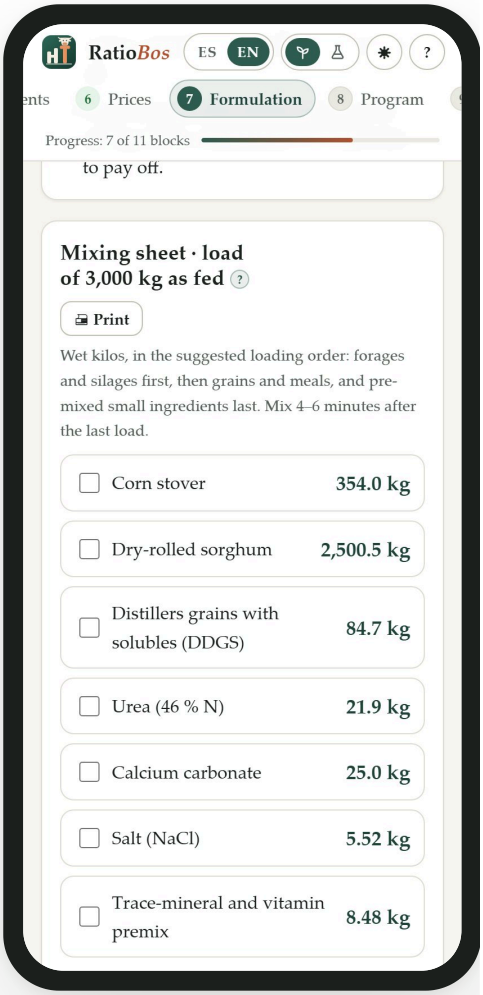


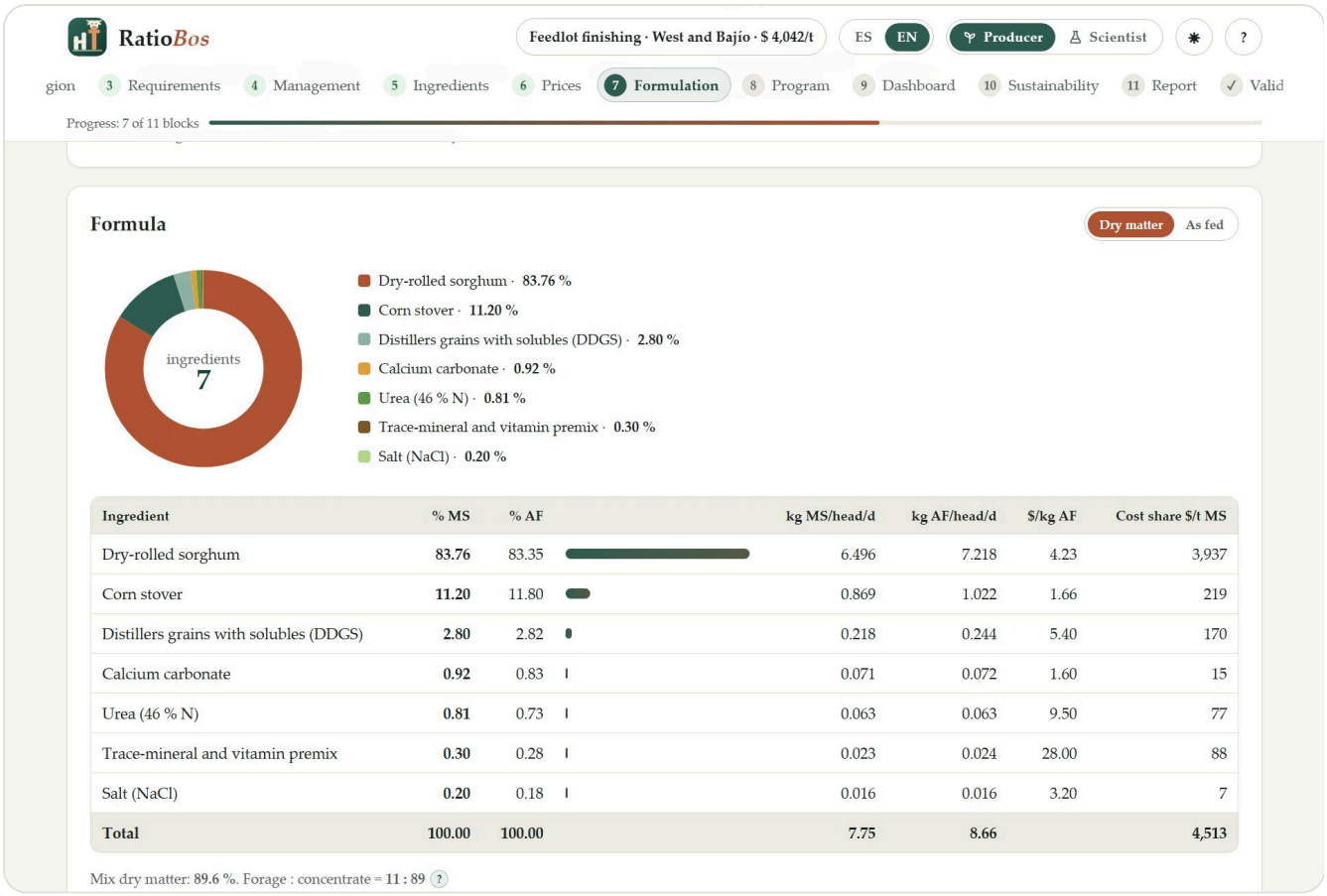

FIGURE 7.5 **On a phone**, by the wagon.

| ORDER | INGREDIENT | KG |
|---|---|---|
| 1 | Corn stover | 354.0 |
| 2 | Dry-rolled sorghum | 2,500.5 |
| 3 | Distillers grains (DDGS) | 84.7 |
| 4 | Urea · limestone · salt · premix *(pre-blended)* | 21.9 · 25.0 · 5.52 · 8.48 |
| | **Load total** · $12,125.09 · feeds 346 head-days | **3,000** |

> ! **CAUTION · UREA, SALT AND PREMIX**
>
> Together they are 61 kg in a 3,000 kg load. Tipped straight into the wagon they spread badly and some animals get too much. **Blend them first with about 50 kg of ground sorghum** and add that premix last. Poorly mixed urea can poison cattle.

## 7.5 The formula and the rumen traffic light

The formula appears as a doughnut and a table, on a dry-matter or as-fed basis (button at the top right). The table also shows how many kilos of each ingredient each animal eats and how much each adds to the cost.




| Ingredient | % MS | % AF | kg MS/head/d | kg AF/head/d | $/kg AF | Cost share $/t MS |
|---|---|---|---|---|---|---|
| Dry-rolled sorghum | **83.76** | 83.35 | 6.496 | 7.218 | 4.23 | 3,937 |
| Corn stover | **11.20** | 11.80 | 0.869 | 1.022 | 1.66 | 219 |
| Distillers grains with solubles (DDGS) | **2.80** | 2.82 | 0.218 | 0.244 | 5.40 | 170 |
| Calcium carbonate | **0.92** | 0.83 | 0.071 | 0.072 | 1.60 | 15 |
| Urea (46 % N) | **0.81** | 0.73 | 0.063 | 0.063 | 9.50 | 77 |
| Trace-mineral and vitamin premix | **0.30** | 0.28 | 0.023 | 0.024 | 28.00 | 88 |
| Salt (NaCl) | **0.20** | 0.18 | 0.016 | 0.016 | 3.20 | 7 |
| **Total** | **100.00** | **100.00** | **7.75** | **8.66** | | **4,513** |

**FIGURE 7.6** **Formula on a dry-matter basis.** Sorghum 83.76 %, stover 11.20 %, DDGS 2.80 %, limestone 0.92 %, urea 0.81 %, premix 0.30 % and salt 0.20 %.

Further down, two bars compare the day's energy and protein against what the animal needs, and the **rumen-health traffic light** checks what least cost might neglect.

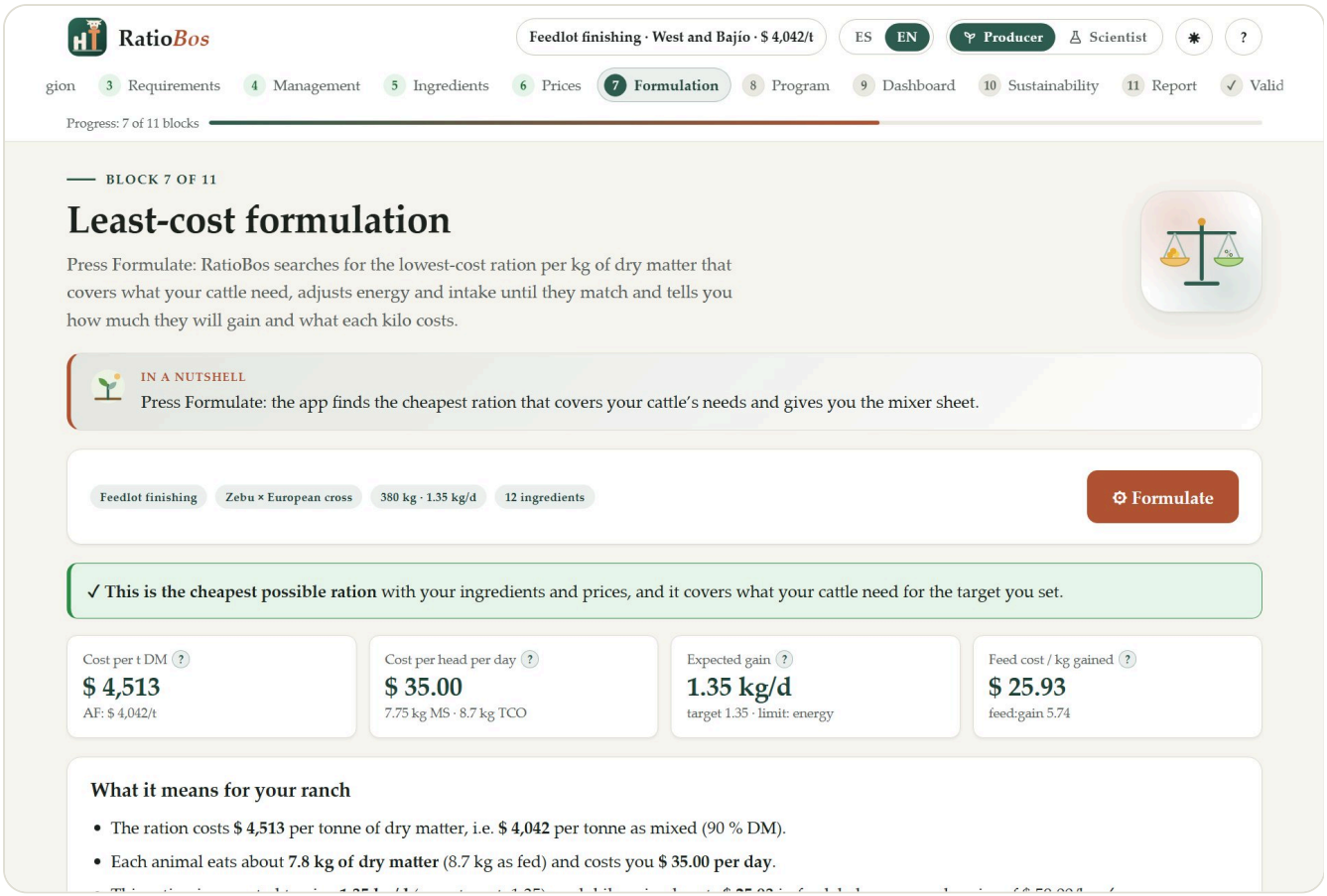



**FIGURE 7.7** **The day's energy and protein, rumen health and footprint.**

→ DECISION RULE · RUMEN TRAFFIC LIGHT

| | |
|---|---|
| **Effective fibre** (peNDF) | Below 7 % in finishing: risk of acidosis and bloat. Add stover. |
| **Total fat** | Above 6–7 % slows fibre digestion. |
| **Urea** | Amber above 1 % of DM or 25 % of the protein; red above 1.2 % or 35 %: toxicity risk. |
| **Sulfur** | Amber above 0.30 %, red above 0.40 %: polioencephalomalacia. The light does not count water; if your water is high in sulfate, stay well below the limit. |
| **Ca : P** | Green from 1.5 : 1; amber between 1.2 and 1.5; red below 1.2. A low Ca : P raises the risk of urinary stones in males: add a little limestone. |

◆ IN THE PEN

All green except **Ca : P**, in amber (1.49 : 1, just under 1.5). In bulls it is worth fixing: if in Block 6 we raise the limestone minimum to `1.1` % and formulate again, Ca : P rises to **1.69 : 1** and a tonne of DM costs only $2 more ($4,515). To keep the example's numbers, the following chapters stay with the original formula.

## 7.6 Which ingredients are worth watching?

The **Unused ingredients · entry price** table tells you at what price an ingredient left out would start to pay off. Use it when negotiating with your supplier.

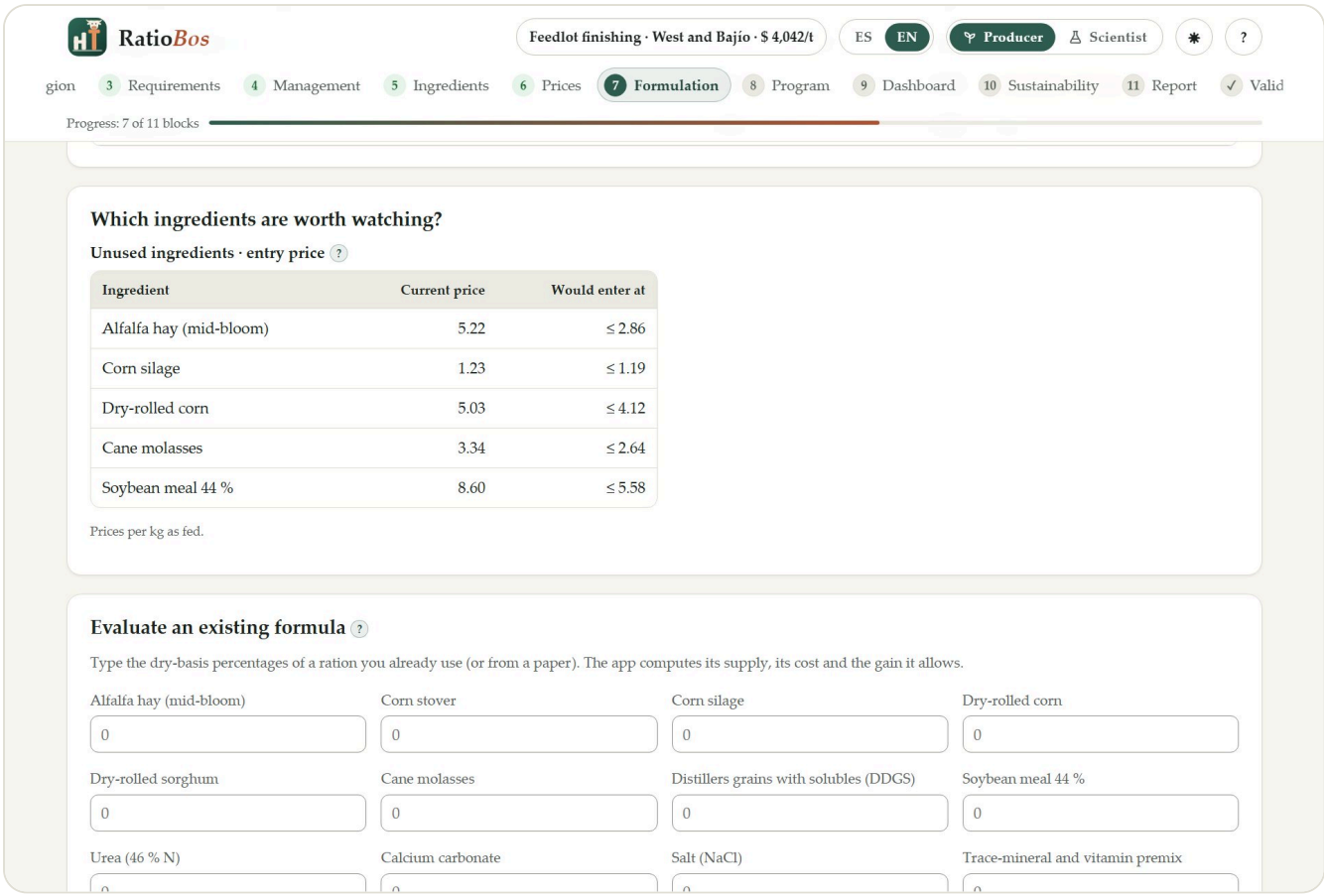



**FIGURE 7.8** **Entry price.** If corn silage drops from $1.23 to $1.19, it enters the ration.

## 7.7 If “No solution” appears

It means no blend of your ingredients meets everything at once. In growing animals, the usual cause is a gain target higher than the ingredients or the climate allow. The app works out the **maximum feasible gain** and offers a button to use it.

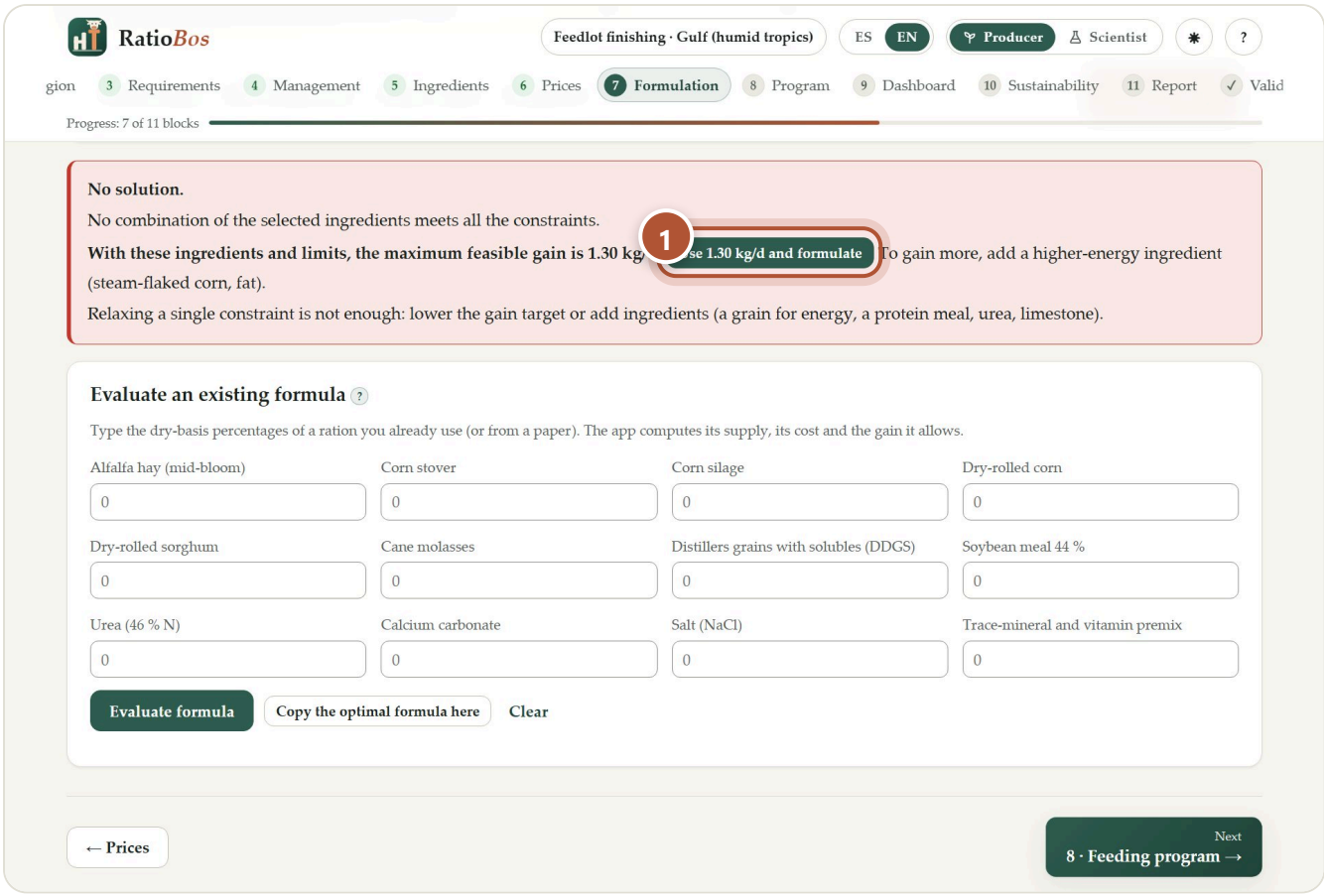



**FIGURE 7.9** **The same pen in Veracruz, dry season.** In that heat the 1.35 kg/d target is not possible: the maximum is 1.30.

1 **Use 1.30 kg/d and formulate**.

### What do I do if there is no solution? Try in this order:

1. **Does the app offer a maximum gain?**
   Use it, or add a higher-energy ingredient (steam-flaked corn, fat) to reach your target.

2. **Did you edit the Block 3 table?**
   Press ↺ *Recompute from the animal* and formulate again.

3. **Does it say "No selected ingredient supplies…"?**
   Tick one in Block 5 that supplies it: limestone for calcium, salt for sodium, a meal for protein.

4. **Still no solution?**
   The app lists the constraints where relaxing **just one** would give a solution. Check those limits in Block 6.

BLOCK 8 · PROGRAM

# Feeding program

8

The whole stay in the pen, day by day: how many days it takes, how much feed and money it uses and what is left per head.

## 8.1 The phases

For a feedlot the app proposes three phases: **receiving and step-up** (plenty of forage so the rumen adapts to grain), **growing** and **finishing**. For each you can change its end weight, gain target and forage range. Enter the **sale weight** and press ⚙ Formulate the program.

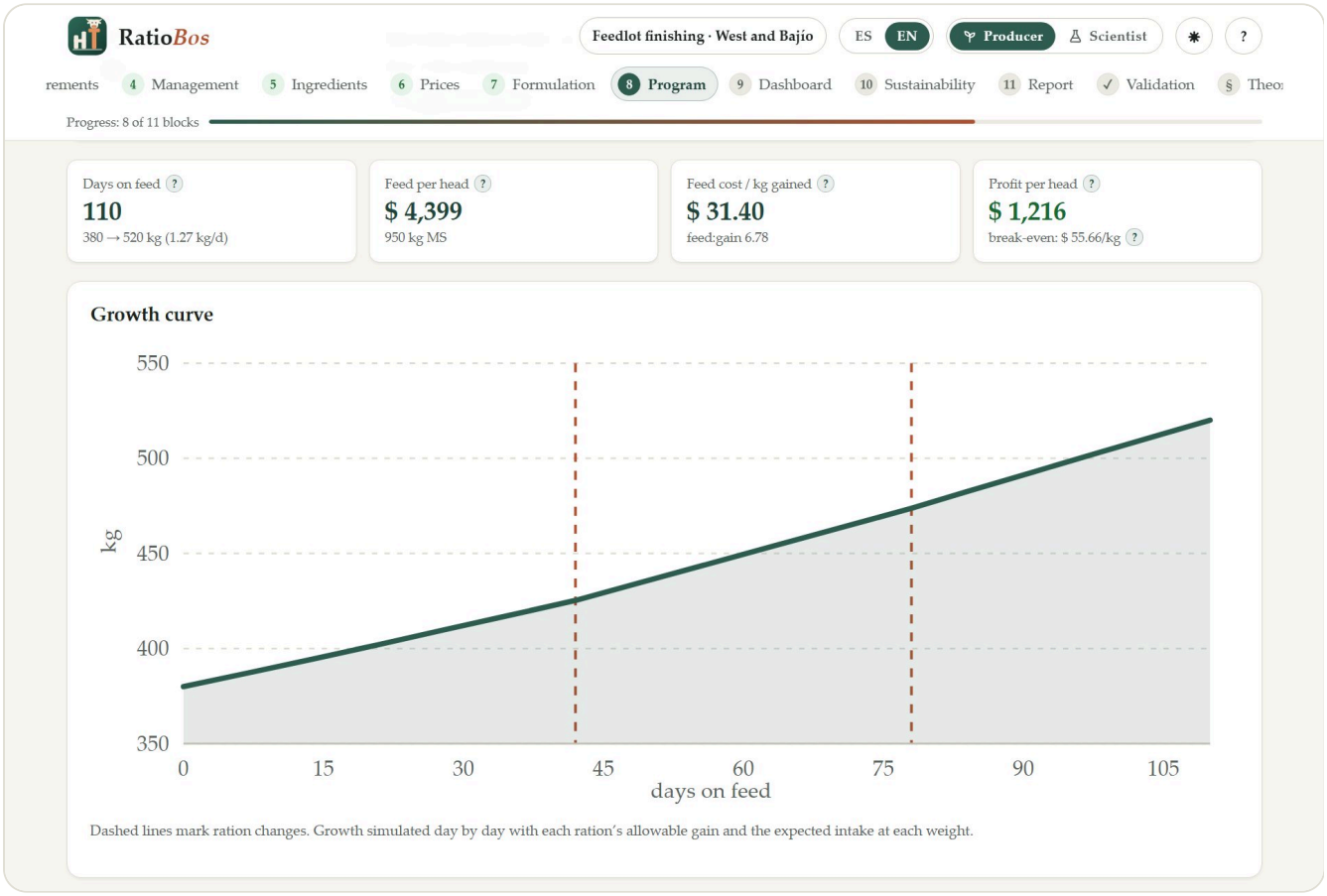



**FIGURE 8.1** **Block 8.** Four cards sum up the whole feeding period.

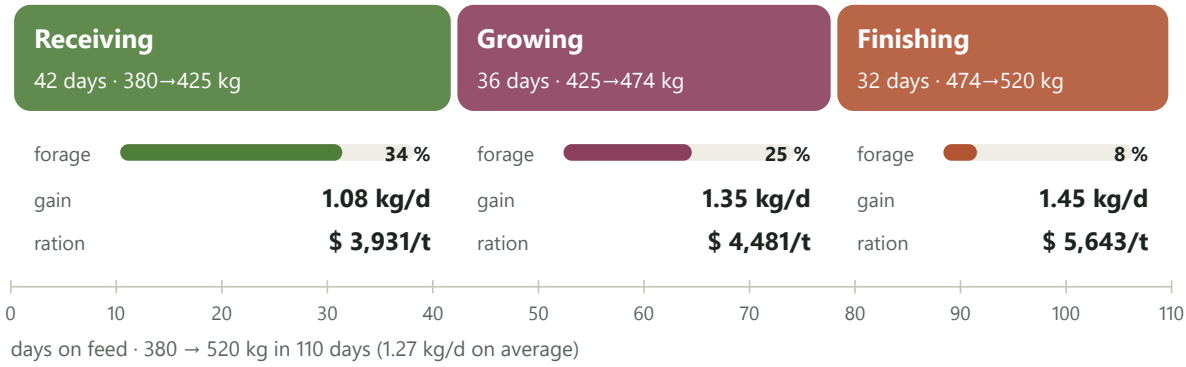



**FIGURE 8.2** **The program at a glance.** As forage drops, gain rises, but so does the ration cost. Data from the case program.

? **WHY IS THE PROGRAM'S GAIN LOWER THAN THE TARGET?**

Because receiving, with 34 % forage, gains less (1.08 kg/d) and because in every phase the app grows the animal at the gain the ration **really** allows, day by day. The 110-day average is 1.27 kg/d. For the same reason the program's cost per kilo ($31.40) is higher than that of the Block 7 ration ($25.93), computed for the 380 kg bull: finishing is dearer and the heavier animal converts worse.

## 8.2 The growth curve

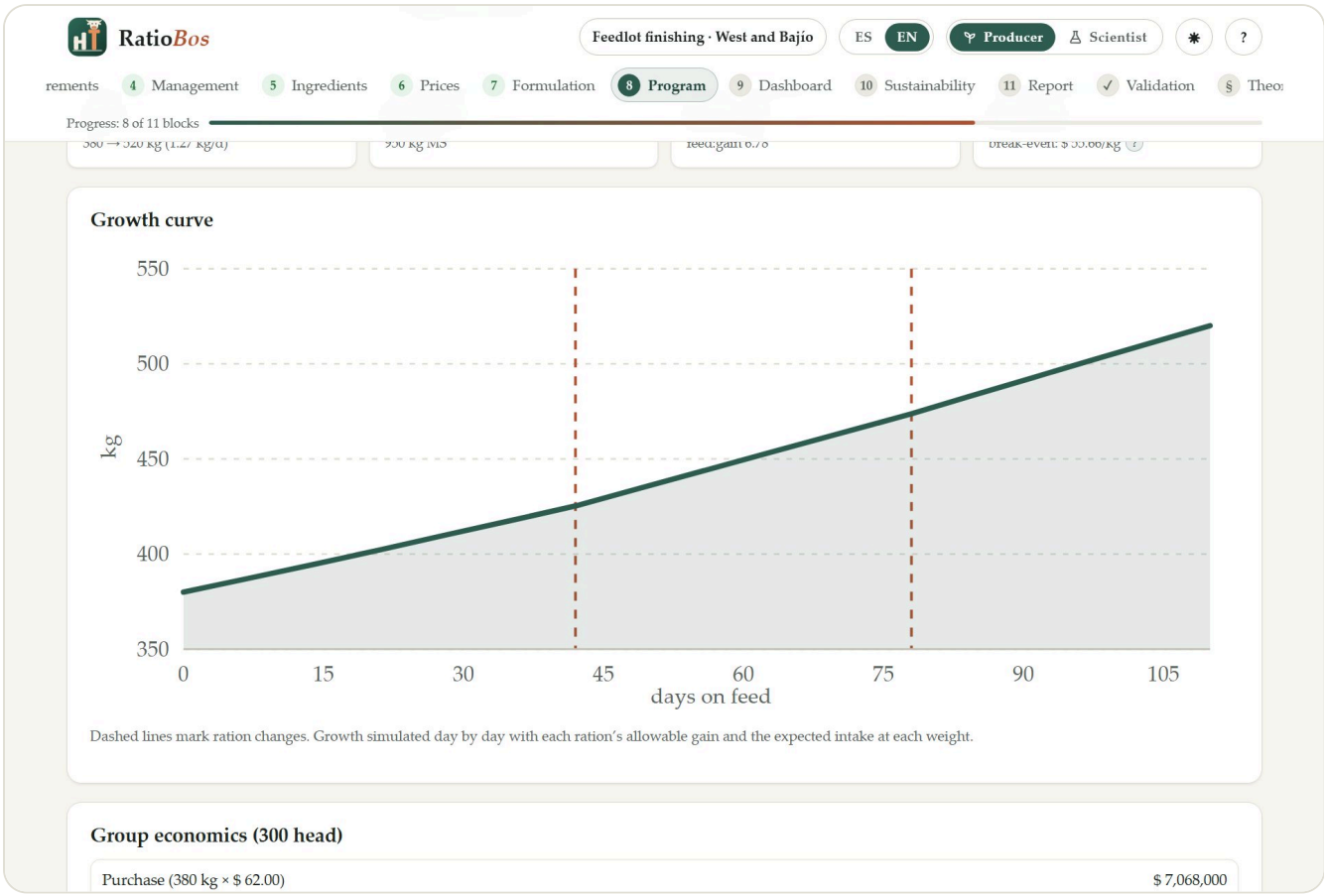



**FIGURE 8.3** **Growth curve.** Dashed lines mark the ration changes.

## 8.3 Batch economics

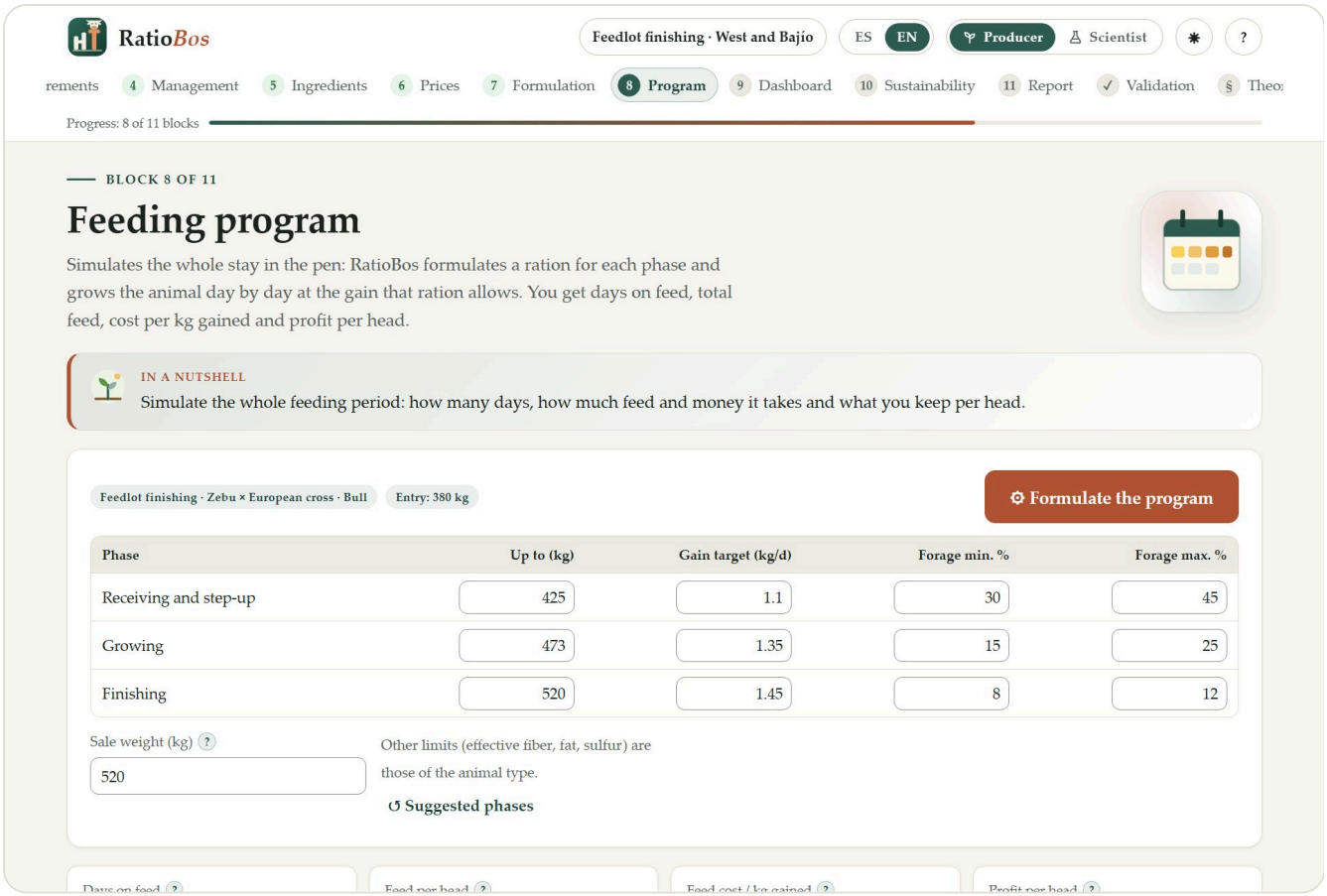



**FIGURE 8.4** **Batch economics and phase summary.**

1 ⬇ Download program (CSV) to open it in Excel.

PURCHASE + FEED + OTHER

**$ 8,684,650**

→

SALE OF 300 BULLS AT 520 KG

**$ 9,049,594**

◆ **IN THE PEN**

The program takes **110 days** from 380 to 520 kg. Each bull eats 950 kg of dry matter, costing **$4,399**. Profit is **$1,216 per head** and **$364,943** for the lot. The **break-even price** is $55.66: if live weight sells below that, the lot loses money. With a margin of only $2.34 per kilo over break-even, it pays to agree the sale before buying.

BLOCK 9 · DASHBOARD

# 9 Results dashboard

Every number that matters in one view and in charts: what the ration achieves, the most profitable gain, the cost in each region and the grain-price risk.

## 9.1 Achieved performance

The table sets the **current ration** beside the **whole program** with the four classic feedlot indicators.

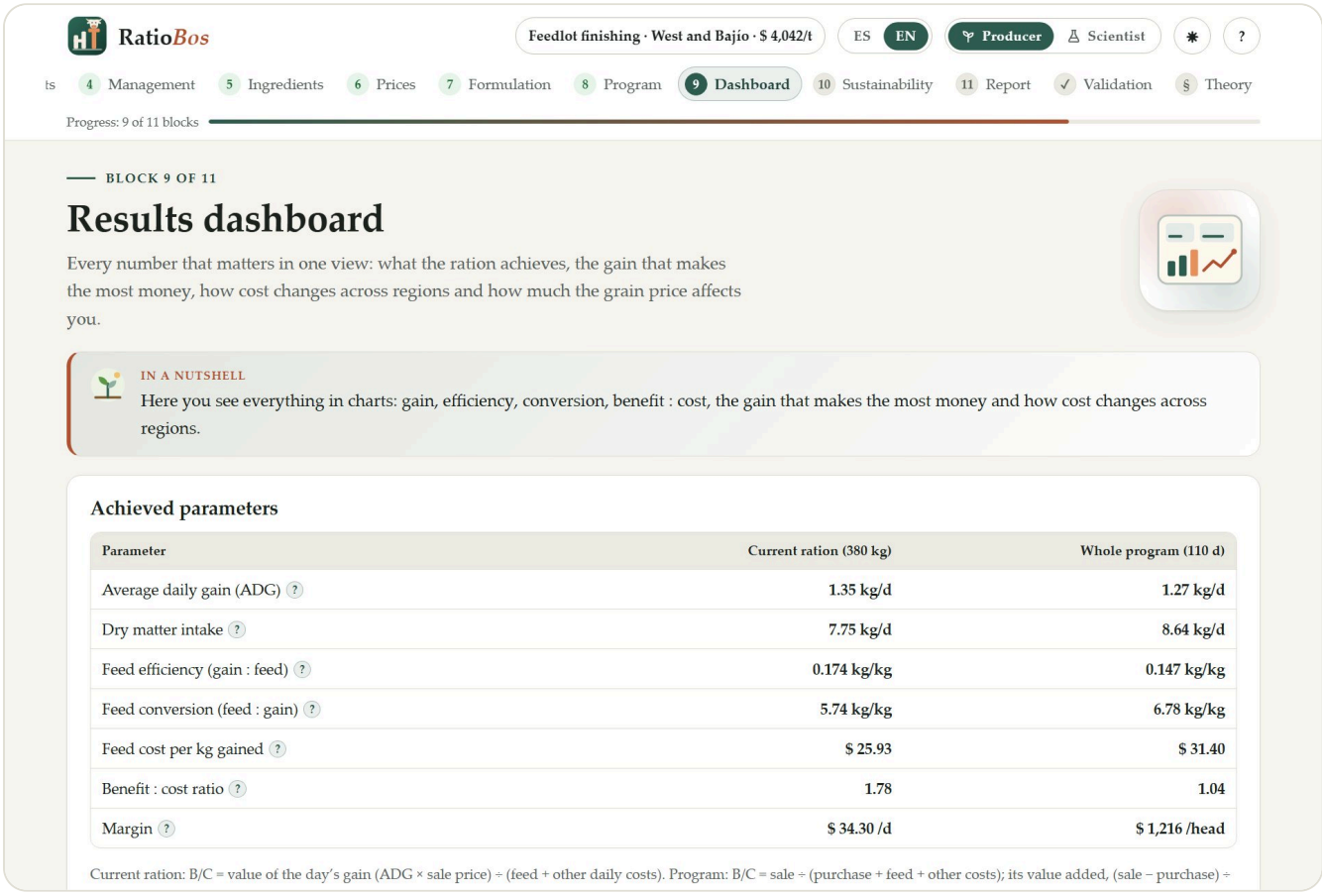



**FIGURE 9.1** **Achieved performance.** Each indicator has its **?** with the decision scale.

| INDICATOR | HOW IT IS COMPUTED | RATION | PROGRAM |
|---|---|---|---|
| **Average daily gain (ADG)** | kg gained ÷ days | 1.35 | 1.27 |
| **Feed efficiency** | gain ÷ feed (higher is better) | 0.174 | 0.147 |
| **Feed conversion** | feed ÷ gain (lower is better) | 5.74 | 6.78 |
| **Benefit : cost** | money in ÷ money out (over 1 = profit) | 1.78 | 1.04 |

? **WHY TWO SUCH DIFFERENT BENEFIT : COST RATIOS?**

The ration ratio compares the value of what the animal gains in a day with what feeding it costs that day: 1.78 says each peso of feed and management produces $1.78 of beef. The program ratio also includes **buying the animal**, the biggest investment: 1.04 is the real return on the whole business. The app also shows the **value added** (sale minus purchase, over feed and other costs): 1.23.

## 9.2 Which gain pays best?

Faster gain does not always pay: each extra kilo demands a dearer ration. The app formulates for several gain targets and plots two curves: the **margin per head per day** (green) and the **feed cost per kilo gained** (terracotta). The circle marks the economic optimum.

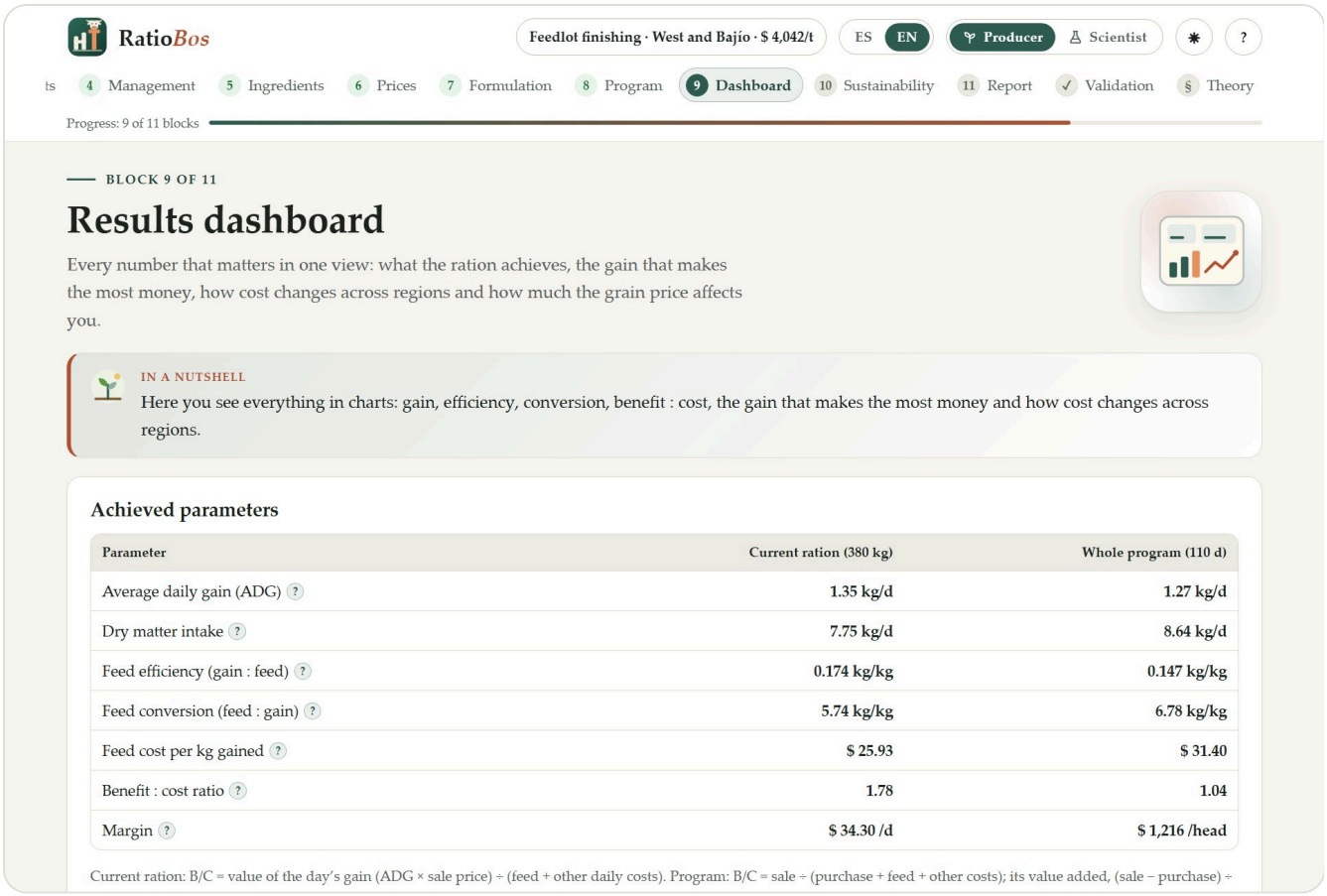



**FIGURE 9.2** **The optimal gain.** The dashed line is your ration.

> **IN THE PEN**
>
> At our prices, the highest daily profit comes at **1.40 kg/d**: $36.44 per head per day against $34.30 at our 1.35 target. The gap is small: the target is already close to the optimum and the app says so with a tick.

## 9.3 The same animal in every region

The app repeats your formulation in all 9 regions, with the season's climate, your shade and each region's estimated prices. If a region's climate does not allow your target, it uses the maximum feasible gain there and marks it with an asterisk.

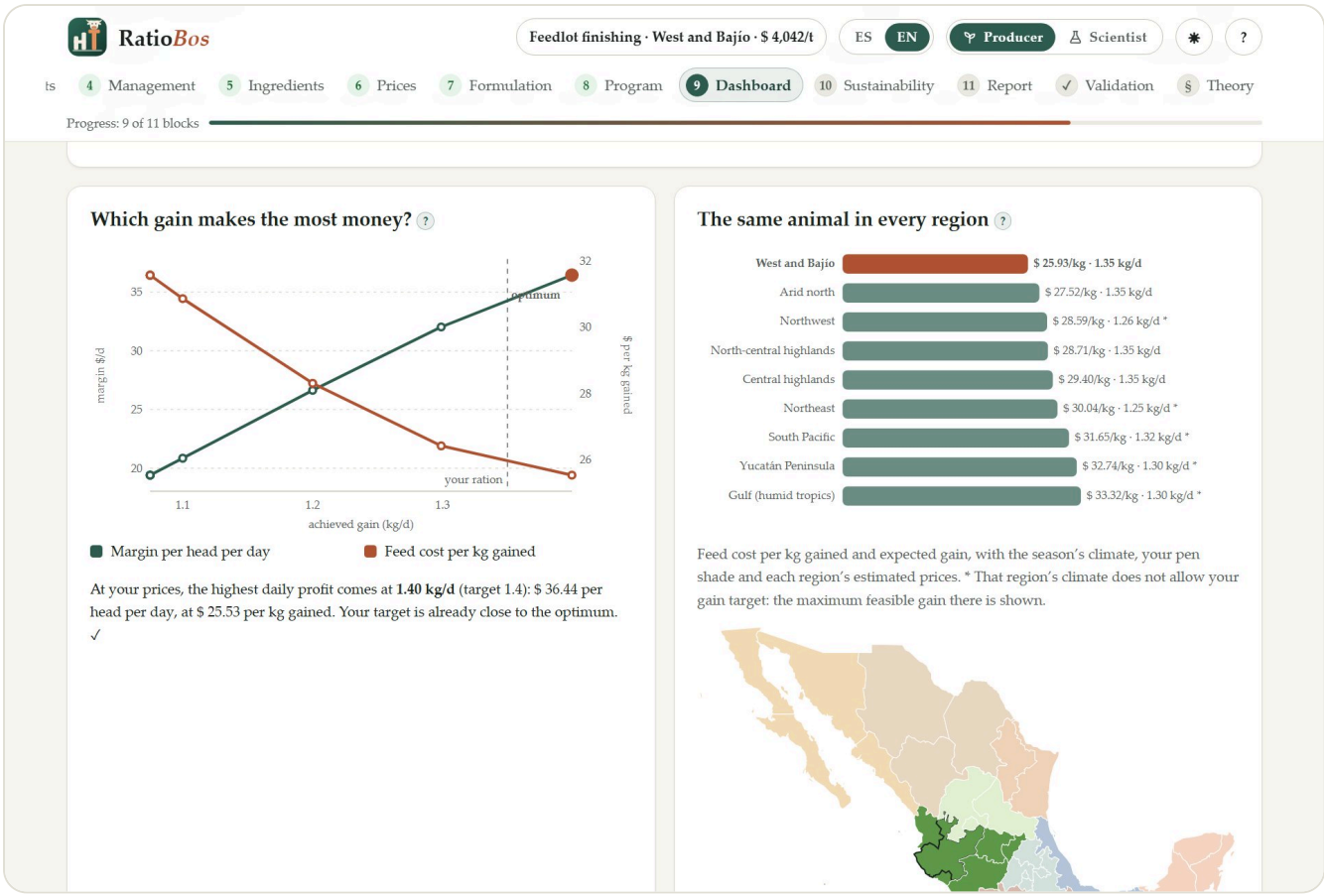



**FIGURE 9.3** **Feed cost per kilo gained in each region**, dry season. Your region is in terracotta.

## 9.4 If the grain price changes

The chart moves the price of the ingredient that weighs most in the cost (sorghum, in our case) from 70 % to 130 % of today's and reformulates at each point. The green line is the cost per tonne; the terracotta one, how much sorghum stays in the ration.

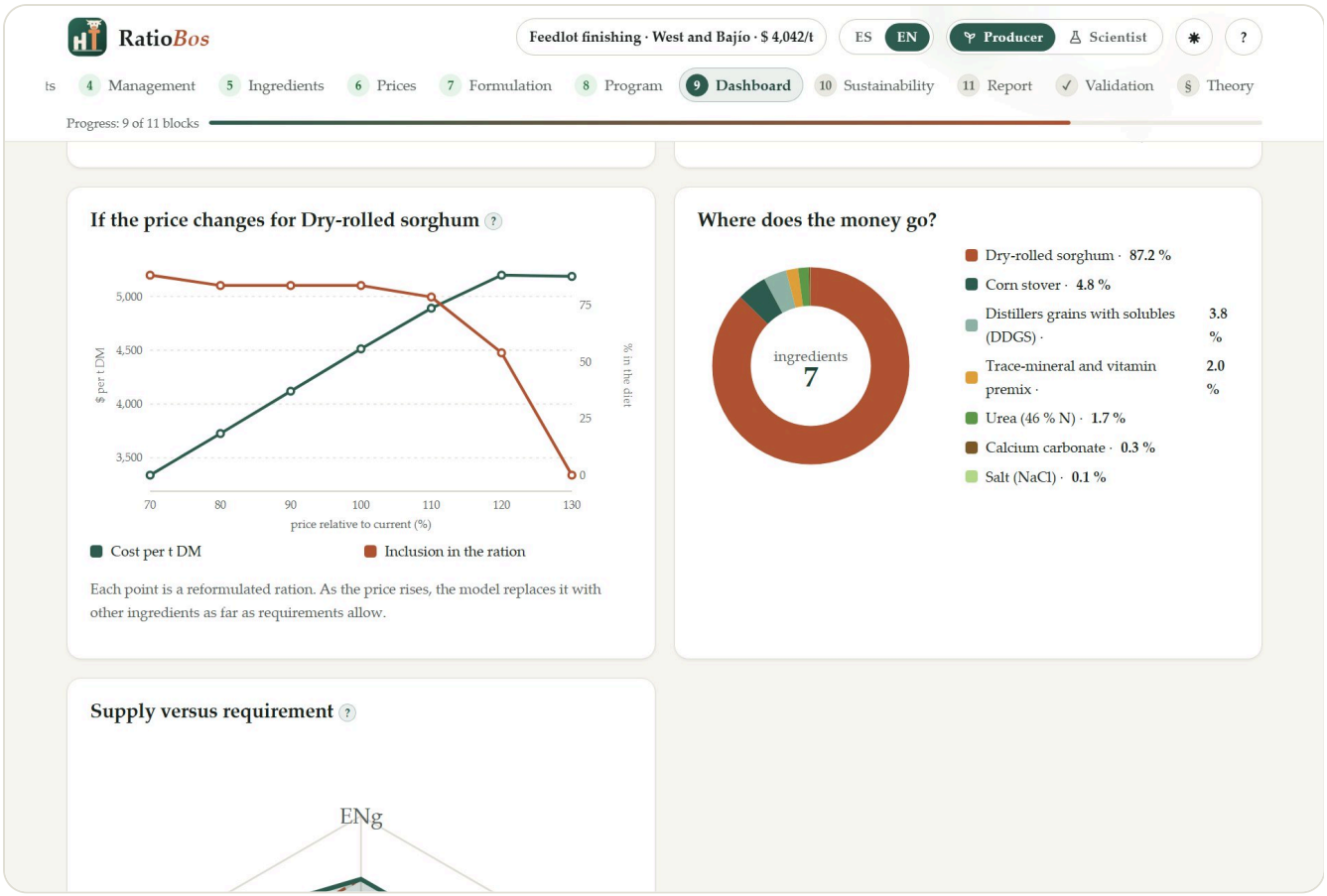



**FIGURE 9.4** **Sorghum-price sensitivity** and where the money goes.

> ✓ **TIP · HOW TO READ IT**
>
> While the terracotta line is flat, sorghum is irreplaceable and its price hits your cost in full: that is the ingredient to buy ahead or under contract. When the line drops, the app has found a substitute. In our case, from a 10 % rise it starts replacing it with silage, molasses and DDGS, and from 25 % it switches entirely to dry-rolled corn.

## 9.5 Compare yourself with published feedlots

At the end of the dashboard is a table with gain, intake, efficiency, conversion and benefit : cost from experimental feedlots published in Mexico and the southern United States, from CIPES in Sonora to Sinaloa, Zacatecas and El Centro, California. Use it to check that your numbers are reasonable.

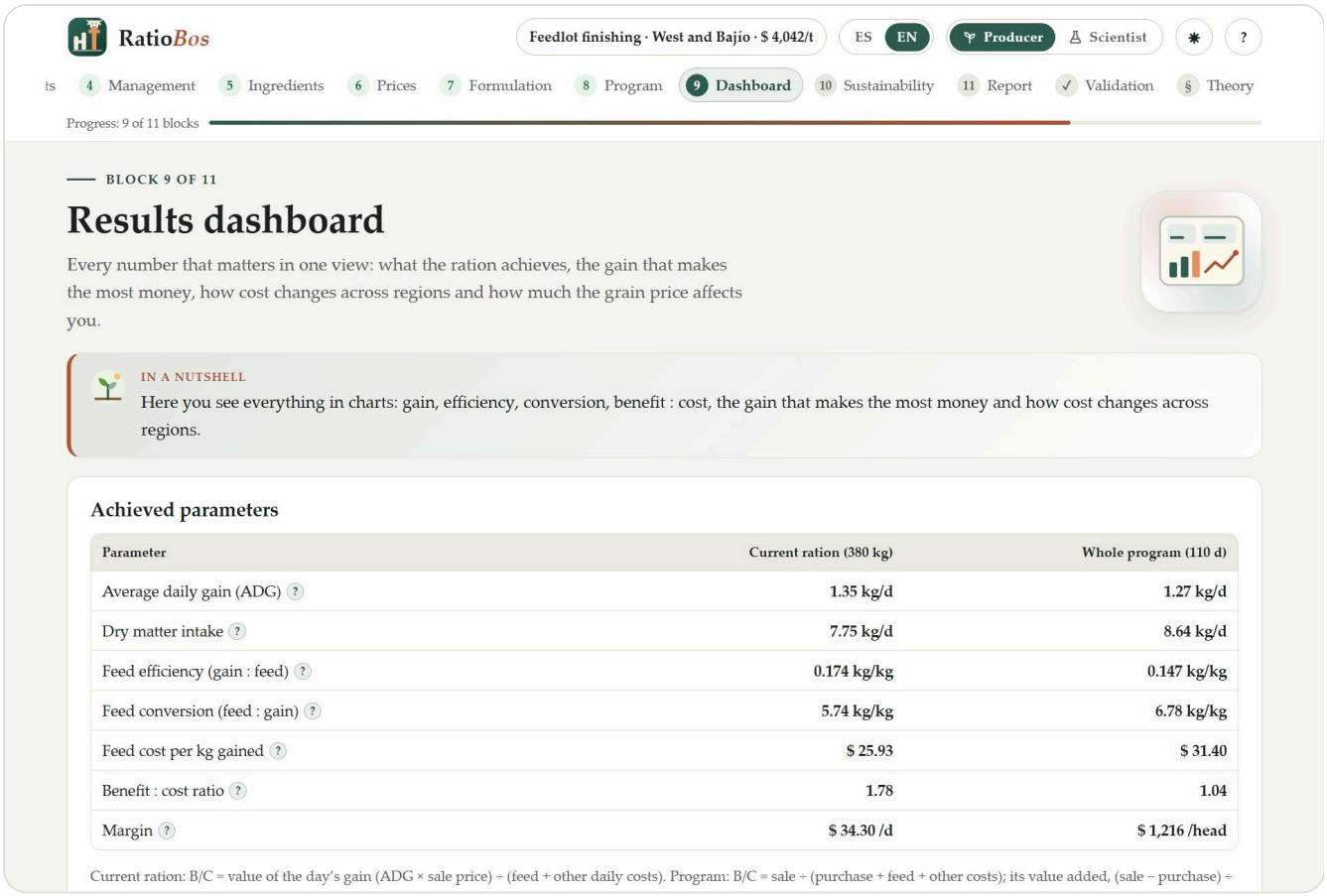



**FIGURE 9.5** **Published performance from reference feedlots.**

BLOCK 10 · SUSTAINABILITY

# Environmental footprint and sustainable production

10

Producing beef uses water and land and leaves gases, nutrients and waste. Here you see how much, and look for the balance between profit and the environment.

RatioBos uses the word **sustainability** as FAO defines it: producing today without compromising the capacity to produce tomorrow, in its three dimensions at once: **environmental**, **economic** and **social**. That is why this block does not seek the "greenest" ration at any price, but the one that cuts the footprint and still pays.

## 10.1 The sustainability traffic light

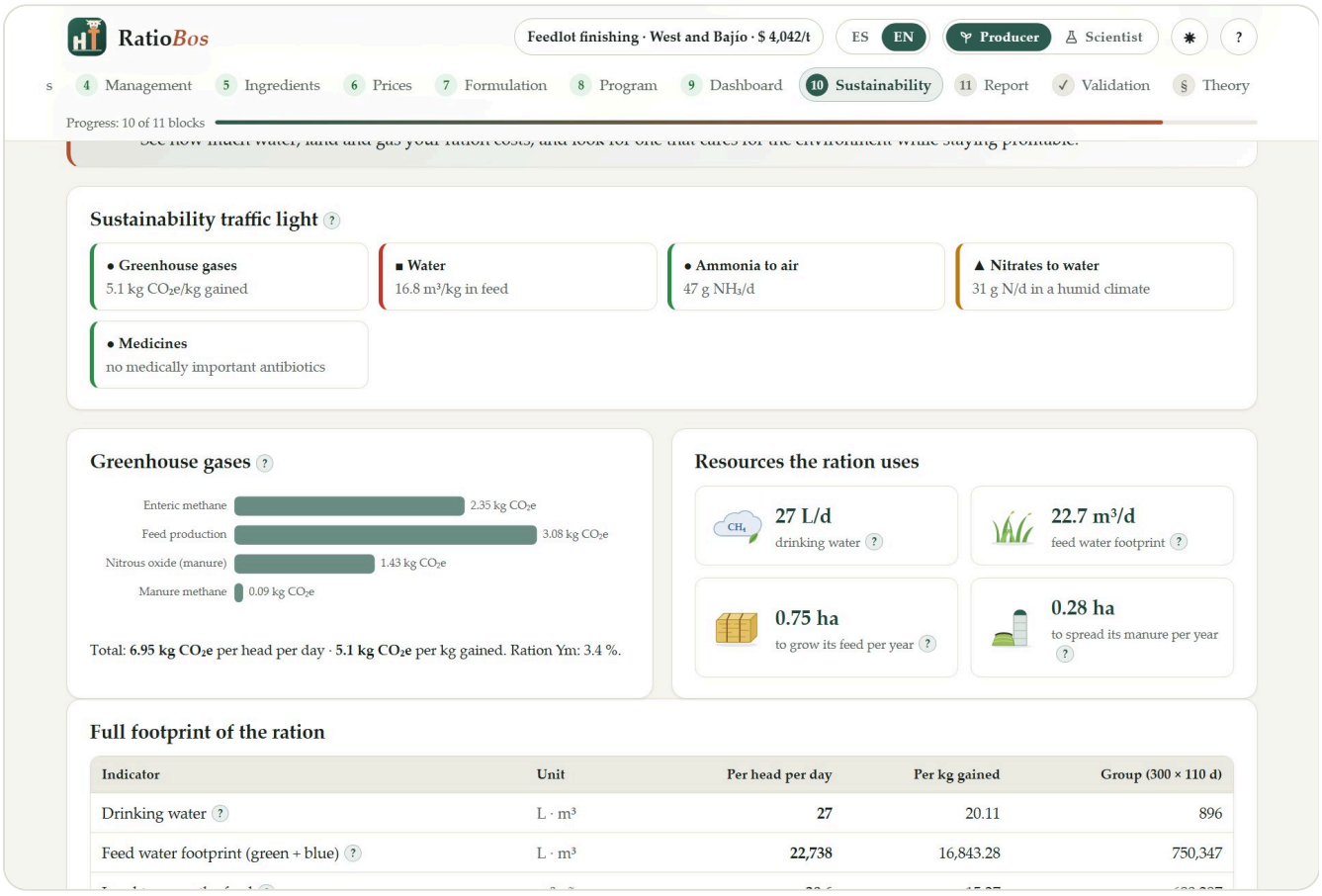



**FIGURE 10.1** **Sustainability traffic light, gases and resources.**

| LIGHT | WHAT IT MEASURES | IN OUR CASE |
|---|---|---|
| **Greenhouse gases** | Methane from the rumen and manure, nitrous oxide and feed production, in $CO_2$ equivalent per kg gained. | 5.1 kg |
| **Water** | Rain and irrigation water it took to grow the feed, per kg gained. | 16.8 $m^3$ |
| **Ammonia to air** | Nitrogen volatilised from manure: odour, fine dust and acid rain. | 47 g/d |
| **Nitrates to water** | Nitrogen that can leach to wells and rivers in a wet climate. | 31 g/d |
| **Medicines** | Whether you use antibiotics important in human medicine (WHO list). | none |

## 10.2 Gases and resources

The gas bars split where each kilo of $CO_2$ equivalent comes from. In our case **feed production** (3.08 kg a day) outweighs **rumen methane** (2.35 kg): on a high-grain ration the rumen makes little methane, but the grain carries its own footprint from the field. Beside it, four cards give drinking water, the feed's water footprint, the land to grow it and the area to spread the manure.

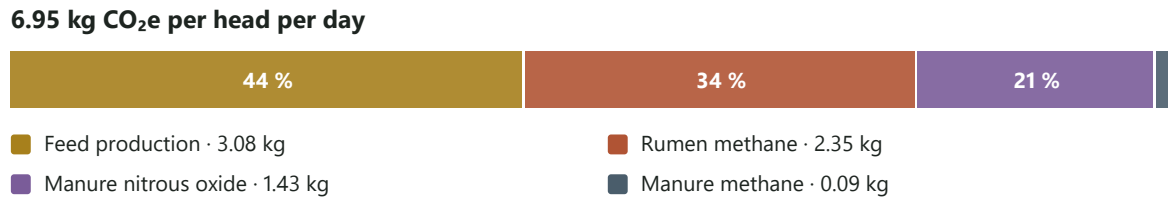
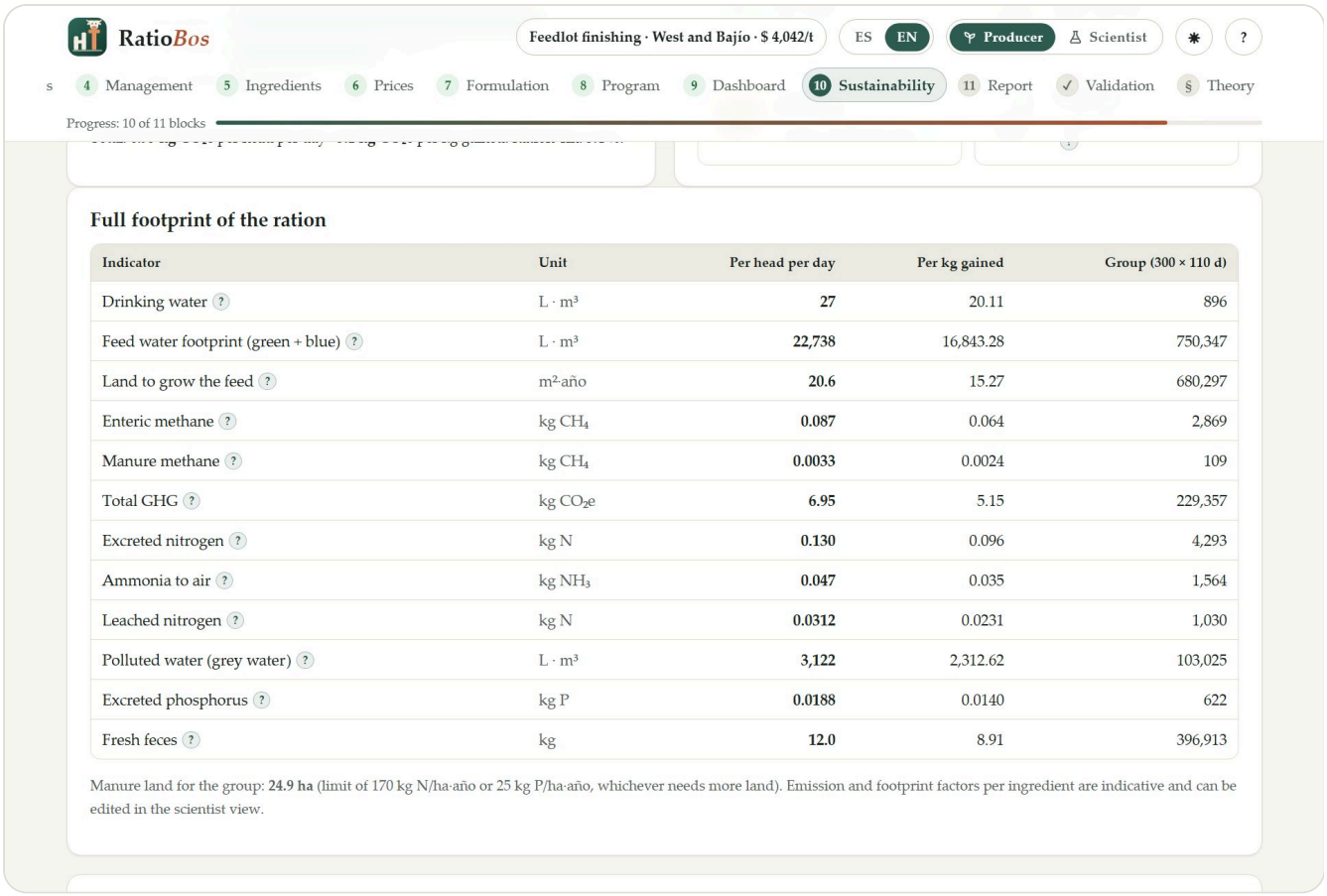



FIGURE 10.2 **Where the gases come from.** On a high-grain ration, the field that grew the feed weighs more than the rumen. Data from the case.




**Full footprint of the ration**

| Indicator | Unit | Per head per day | Per kg gained | Group (300 × 110 d) |
|---|---|---|---|---|
| Drinking water | L · m³ | **27** | 20.11 | 896 |
| Feed water footprint (green + blue) | L · m³ | **22,738** | 16,843.28 | 750,347 |
| Land to grow the feed | m²·año | **20.6** | 15.27 | 680,297 |
| Enteric methane | kg $CH_4$ | **0.087** | 0.064 | 2,869 |
| Manure methane | kg $CH_4$ | **0.0033** | 0.0024 | 109 |
| Total GHG | kg $CO_2e$ | **6.95** | 5.15 | 229,357 |
| Excreted nitrogen | kg N | **0.130** | 0.096 | 4,293 |
| Ammonia to air | kg $NH_3$ | **0.047** | 0.035 | 1,564 |
| Leached nitrogen | kg N | **0.0312** | 0.0231 | 1,030 |
| Polluted water (grey water) | L · m³ | **3,122** | 2,312.62 | 103,025 |
| Excreted phosphorus | kg P | **0.0188** | 0.0140 | 622 |
| Fresh feces | kg | **12.0** | 8.91 | 396,913 |

Manure land for the group: **24.9 ha** (limit of 170 kg N/ha·año or 25 kg P/ha·año, whichever needs more land). Emission and footprint factors per ingredient are indicative and can be edited in the scientist view.

**FIGURE 10.3** **Full footprint** per head per day, per kg gained and for the whole lot.

◆ **IN THE PEN**

Our 300 bulls, over 110 days, produce about **397 tonnes of fresh faeces** with 4.3 t of nitrogen and 0.6 t of phosphorus. To spread it without polluting takes **24.9 hectares** of cropland (limit of 170 kg N or 25 kg P per hectare a year). If you do not have them, compost the manure or sell it to someone who does.

## 10.3 Medicines by category

From what you entered in Block 4, the app works out how much of each product each animal and the whole lot receive, and classifies it: ionophores (no importance in human medicine), hormonal implants, antiparasitics, vaccines and, if used, antibiotics by WHO category and β-agonists.

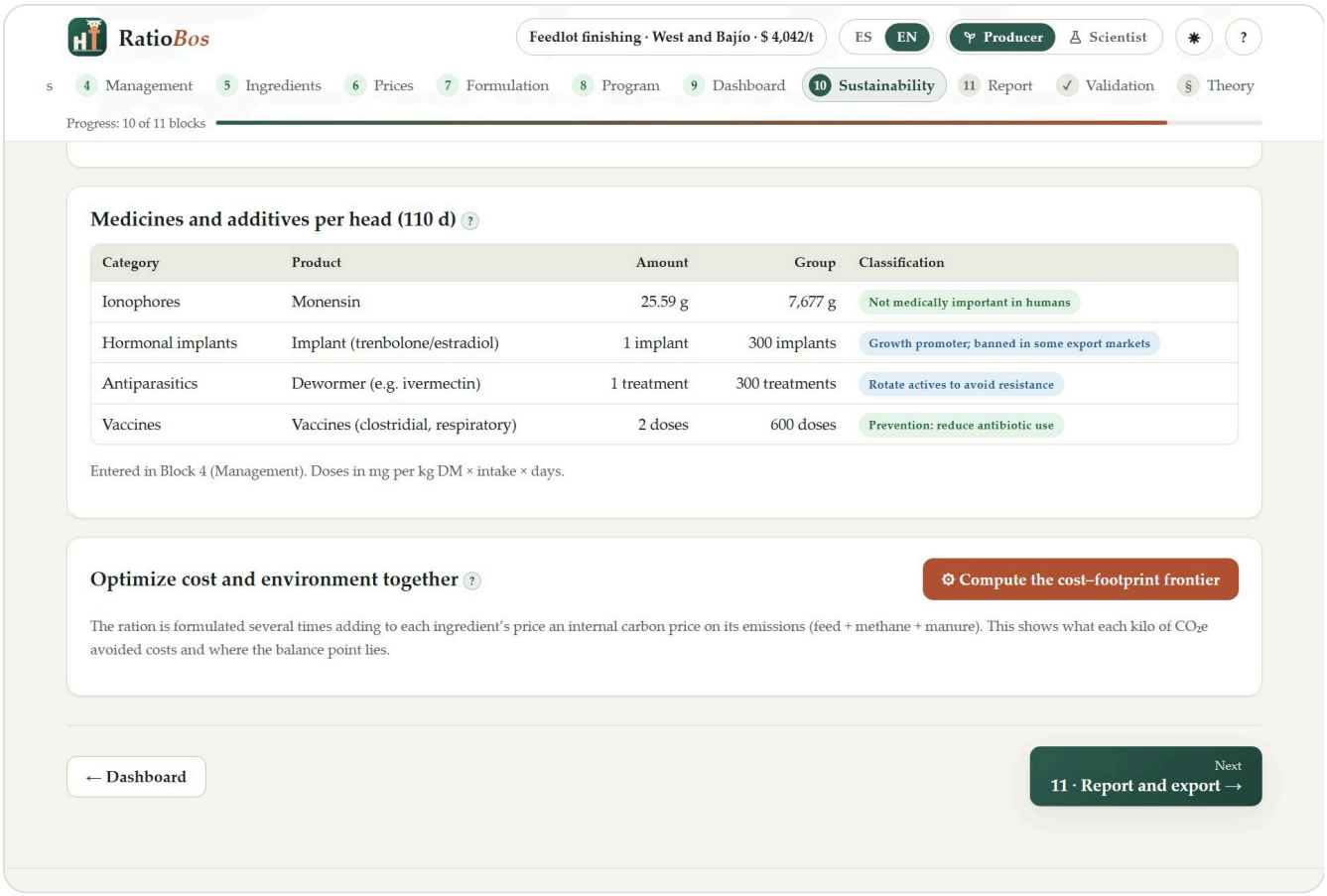



**FIGURE 10.4** **Medicines and additives per animal**, 110 days.

## 10.4 Optimising cost and environment together

The **Compute the cost–footprint frontier** button formulates several times, adding to each ingredient's price an **internal carbon price** for its emissions, from $0 to $20,000 per tonne of $CO_2e$. The result is a curve of the best possible rations; no other is both cheaper and cleaner. The **balance** point is the one that removes the most footprint per extra peso.




| Carbon price $/t $CO_2e$ | $ per kg gained | kg $CO_2e$ per kg gained | Cost to avoid 1 t $CO_2e$ |
|---|---|---|---|
| 0 | 25.93 | 5.15 | — |
| 2,000 | 26.02 | 5.11 | $ 2,814 |
| **5,000** | **26.58** | **4.96** | **$ 3,537** |

**FIGURE 10.5** **Cost–footprint frontier.**

1 **Use the balance ration** sends it to Block 7 as your ration.

LEAST COST

**$ 25.93 · 5.15 kg**

→

BALANCE · $5,000/T $CO_2E$

**$ 26.58 · 4.96 kg**

◆ **IN THE PEN**

With an internal price of $5,000 per tonne, the app swaps part of the sorghum for molasses and DDGS: the footprint falls **3.6 %** (from 5.15 to 4.96 kg $CO_2e$ per kilo gained) and each kilo gained costs **2.5 %** more ($26.58). Each tonne of $CO_2e$ avoided costs about $3,500, in the range of the social cost of carbon. It is a business decision: worth it if your buyer pays a premium for lower-footprint beef.

BLOCK 11 · REPORT

# 11 Report and export

A report for your advisor, tables for Excel and a file with your whole project.

## 11.1 The report

A one-sentence summary and the formula on a dry-matter basis, as fed and in kilos per tonne of mix, with the pen, ranch and date. **Print / PDF** prints it or saves it as a PDF to send by WhatsApp or e-mail.

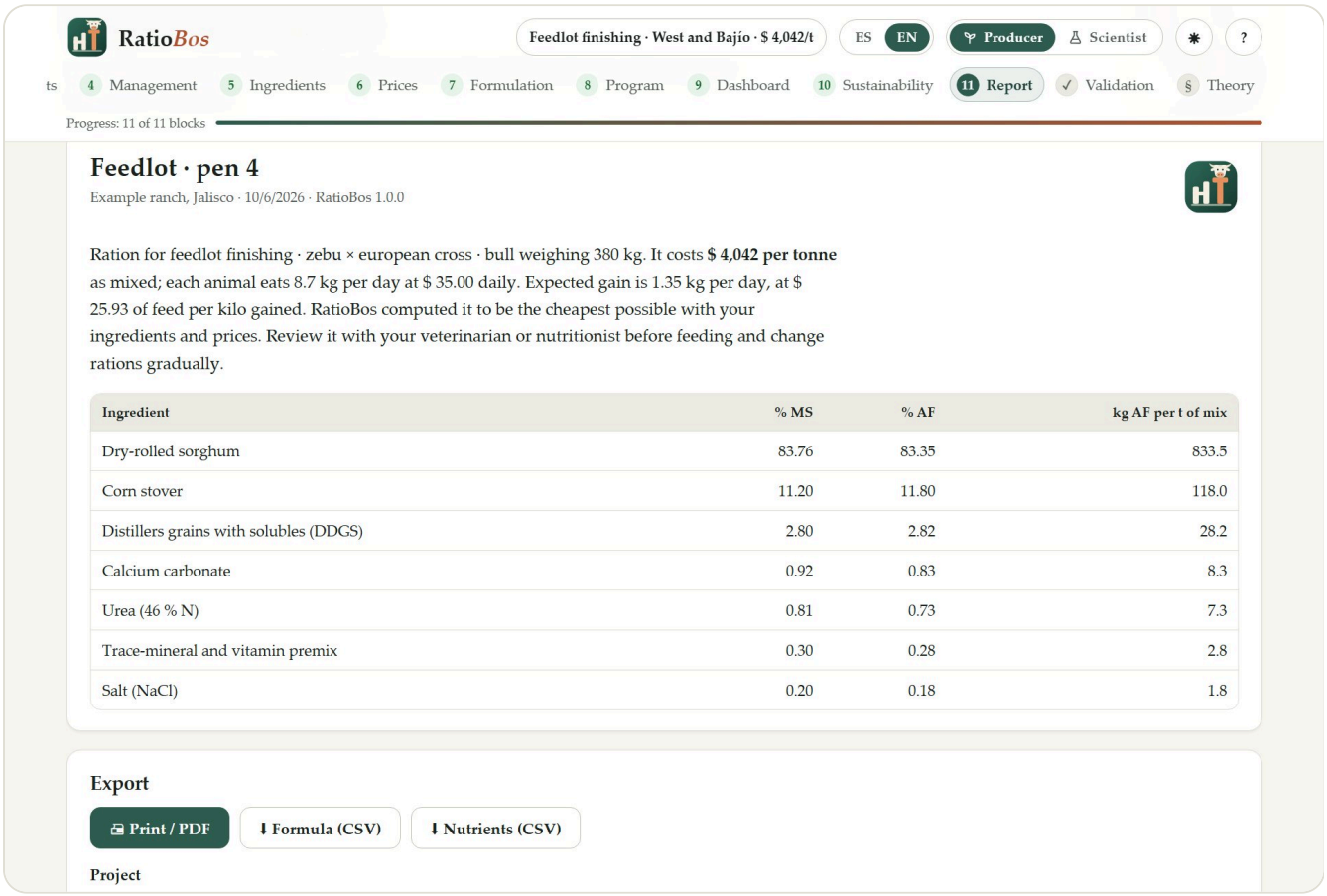



**FIGURE 11.1** Ration report.

## 11.2 Export and save

| BUTTON | WHAT IT DOES |
|---|---|
| ⬇ Formula (CSV) | The formula with prices and cost share, for Excel. |
| ⬇ Nutrients (CSV) | Each nutrient's supply against its minimum and maximum. |
| 💾 Save project (.json) | Everything: animal, region, ingredients, prices, result and program. Open it on another computer with 📂 Open project. |
| Reset everything | Back to factory values. Save your project first. |

✓ **TIP · ONE FILE PER PEN AND DATE**

Name the project with the pen and date (for example, "Pen 4 · 2026-10-06"). When the feeding period ends you can compare what the app predicted with what your animals weighed: the best way to fine-tune your ranch's data.

REFERENCE

# From paper to feed bunk

A

The best ration is worthless if it is mixed badly or changed overnight. Five minutes of checking prevent a wrong load and an acidosis outbreak.

## A.1 Before loading the wagon

- [ ] The **prices** in Block 6 are this week's, delivered to the ranch.
- [ ] The silage **dry matter** is up to date: it changes with every silo and with rain. A microwave or oven test is enough.
- [ ] The **weight** and **target** in Block 3 are those of the lot that will eat this ration.
- [ ] Every ingredient in the formula is **in stock**, in the amount needed.
- [ ] The **wagon load** in Block 4 matches your mixer, and the scale is calibrated.
- [ ] Urea, salt, minerals and premix are **pre-blended** with a little ground grain.
- [ ] Your **veterinarian** or nutritionist has reviewed the report. RatioBos is a support tool, not a substitute.

## A.2 Change rations gradually

The rumen needs time to shift its microbes from forage to grain. That is why the program starts with a receiving ration of 30–45 % forage. Between one ration and the next, switch over **3 to 5 days**, blending both or raising grain step by step, and never raise grain by more than 10 points at once.

> ! **CAUTION · SIGNS OF ACIDOSIS**
>
> Loose, bubbly faeces, animals going off feed, lameness a few days later and intake that swings up and down. If you see them, go back to the previous ration and call your veterinarian.

## A.3 Read the bunk every morning

Actual intake rules. Before the first feeding, look at what is left and adjust the day's amount:

→ **DECISION RULE · BUNK READING**

| | |
|---|---|
| **Licked clean** | Raise the day’s amount 5 %. |
| Crumbs, a thin layer | Right. No change. |
| 5–10 % left | Cut 5 %. Check for heat or a sick animal. |
| **Over 10 % left** | Cut 10 % and check the mix, water and pen health. |

✓ **TIP · CLOSE THE LOOP**

Once you know your lot’s actual intake over a week or two, enter it in Block 3 (**Actual DM intake**). The app will use it instead of its prediction, and the expected gain will match your pen.

REFERENCE

# B Troubleshooting

The most common questions, with cause and fix.

**I changed a price and the result disappeared.**

**“No solution” appears.**

**The expected gain is higher than my target.**

**The ration leaves out my main ingredient.**

**The wagon kilos are not what I expected.**

**I opened the app on another computer and my data is missing.**

**The THI says “emergency” but my cattle are fine.**

**Can I formulate for cows on rangeland?**

REFERENCE

# C Glossary

The technical words in this guide, in plain language.

| | |
|---|---|
| **Rumen acidosis**<br>*Acidosis ruminal* | Too much acid in the rumen from much grain and little fibre; cuts intake and causes lameness. |
| **Dry basis (DM)**<br>*Base seca (MS)* | Feed without its water. Ingredients are compared and the ration computed this way. |
| **Benefit : cost**<br>*Beneficio : costo* | Money in over money out. Above 1 means profit. |
| **Biotype**<br>*Biotipo* | Group of breeds that behave alike: British, Continental, zebu, cross, Criollo, dairy. |
| **Feed conversion**<br>*Conversión alimenticia* | Kilos of feed (DM) per kilo of weight gained. Lower is better. |
| **$CO_2$ equivalent ($CO_2e$)**<br>*$CO_2$ equivalente ($CO_2e$)* | Gases added by their climate effect: 1 kg methane = 27 kg $CO_2$; 1 kg nitrous oxide = 273. |
| **Feed efficiency**<br>*Eficiencia alimenticia* | Kilos gained per kilo of feed. Higher is better. |
| **Net energy for gain (NEg)**<br>*Energía neta de ganancia (ENg)* | The feed energy that becomes kilos, in Mcal per kg DM. |
| **Net energy for maintenance (NEm)**<br>*Energía neta de mantenimiento (ENm)* | The energy the animal spends just staying alive, without gaining weight. |
| **Physically effective NDF (peNDF)**<br>*FDN físicamente efectiva (FDNfe)* | Long fibre that makes cattle chew cud and produce saliva, the rumen's natural buffer. |
| **ADG**<br>*GDP* | Average daily gain, in kg per day. |

| | |
|---|---|
| **Ionophore**<br>*Ionóforo* | Additive (monensin, lasalocid) that improves energy use and reduces acidosis. |
| **THI**<br>*ITH* | Temperature-humidity index: the heat the animal feels. |
| **Least cost**<br>*Mínimo costo* | The cheapest ration among all those that meet the requirements. |
| **Entry price**<br>*Precio de entrada* | Price at which an ingredient left out would start to pay off. |
| **Break-even price**<br>*Precio de equilibrio* | Sale price per kilo live weight at which the lot neither gains nor loses. |
| **Premix**<br>*Premezcla* | A pre-blend of the small ingredients with a little ground grain, so they spread evenly. |
| **Metabolizable protein (MP)**<br>*Proteína metabolizable (PM)* | The protein that actually reaches the gut: rumen microbial protein plus undegraded protein. |
| **Sustainability**<br>*Sostenibilidad* | As FAO defines it: producing today without compromising the capacity to produce tomorrow, environmentally, economically and socially. |
| **As fed (AF)**<br>*Tal como se ofrece (TCO)* | Feed with its water, as the wagon scale weighs it. |
| **Urea**<br>*Urea* | Cheap non-protein nitrogen that rumen microbes turn into protein. At most ≈ 1 % of DM. |

PRINT IT AND PIN IT BY THE MIXER WAGON

# RatioBos on one page

hectortecumshe-ai.github.io/RatioBos

Scan to open the app

1 **Animal**
Animal type, biotype (cross, zebu…) and sex.

2 **Region**
Tap your state and season. Check THI, shade and floor.

3 **Requirements**
Lot weight and daily gain target.

4 **Management**
Implant, ionophore, head count, wagon load, buy and sell.

5 **Ingredients**
Tick what you have. Fix the silage DM.

6 **Prices**
This week's delivered price. Check the maximums.

7 **Formulation**
⚙ Formulate. Check the traffic light and print the wagon sheet.

8 **Program**
⚙ Formulate the program: days on feed and profit per head.

9 **Dashboard**
ADG, efficiency, conversion, B : C and the optimal gain.

10 **Sustainability**
Ration footprint and cost–footprint frontier.

11 **Report**
PDF for your advisor and Save project (.json).

## IF "NO SOLUTION"

- Use the maximum gain it proposes.
- Recompute the Block 3 table.
- Tick the missing ingredient.

## BEFORE MIXING

- Prices and silage DM up to date.
- Urea and minerals pre-blended.
- Ration change over 3–5 days.

REFERENCE

# How to cite

If you use RatioBos or this guide in a paper, thesis or report, cite them as follows (APA 7th ed.).

## The app

Mojica-Zárate, H. T., & Barrera-Guzmán, L. Á. (2026). *RatioBos: Guided least-cost diet formulation for beef cattle* (Version 1.0.0) [Software]. Zenodo. https://doi.org/10.5281/zenodo.23197134

## This guide

Mojica-Zárate, H. T., & Barrera-Guzmán, L. Á. (2026). *RatioBos: Producer Guide* (1st ed.). https://hectortecumshe-ai.github.io/RatioBos/instructivo-productor.html

**RatioBos · Producer Guide**, 1st edition, October 2026. Covers RatioBos 1.0.0. Every screenshot comes from the real app with the case data in section 0.5; prices are illustrative.

Decision-support tool: results should be reviewed by a qualified animal nutritionist. © 2026 Héctor Tecumshé Mojica-Zárate. The software is distributed under the MIT License.